EPILEPSY DIGITAL TWIN REPORT · INTEGRATED NEUROLOGICAL DECISION SUPPORT

Neuro Digital Twin Integrated Clinical Report

An integrated clinical assessment combining follow-up visits, seizure history, EEG, ECG, CT/MRI imaging, laboratory results, and pharmacogenomic testing for Focal impaired-awareness seizures.

Patient & Clinical Context

11/11 data categories included · generated 2026-10-05T15:09:53

PATIENT	AGE / SEX	PRIMARY DIAGNOSIS	DIAGNOSED SINCE
John Doe Patient ID SYN-100482	**34 years · Male**	**Focal impaired-awareness seizures** Secondary diagnosis: Secondary generalized tonic-clonic seizures	**12 May 2021**

CURRENT CLINICAL STATUS
Stable outpatient epilepsy state with persistent breakthrough seizure risk

CURRENT THERAPY
Levetiracetam (Keppra) 1000 mg BID · Lamotrigine 150 mg BID

REPORTING PERIOD
Data reviewed: 01 Jan – 30 Jun 2026 (~6 months)
Clinic visit notes cover 01 Jan 2026 to 02 Mar 2026; laboratory, imaging, EEG/ECG, and questionnaire data continue through 30 Jun 2026.

5	12	6	6	12	35	12
Clinical Visits	CBC Reports	CT Studies	MRI Series	ECG Sessions	EEG Recordings	Questionnaires

01 Executive Summary

Across five clinical-note visits from 2026-01-01 to 2026-03-02, the patient has chronic focal epilepsy with persistent breakthrough seizures on levetiracetam and lamotrigine. The clinician repeatedly documented reasonably stable seizure control and favorable quality-of-life trajectory, while seizure burden, sleep quality, trigger exposure, aura symptoms, and medication tolerability continued to fluctuate.

67%	Moderate	Elevated	10 (7 high)
Overall Health Score	Overall Clinical Risk	EEG-Derived Seizure Risk	Genetic Safety Flags

Domain the patient rates worst on questionnaires: Epileptic Triggers. See the Digital Twin Risk Dashboard (Section 13) for the full risk breakdown.

02 6-Month Timeline

01 JAN 2026
Routine epilepsy follow-up with stable clinician impression — Contributes to longitudinal assessment of seizure burden, triggers, sleep, and tolerability.

16 JAN 2026
Seizure clustering documented during follow-up — Raises near-term seizure-monitoring priority despite otherwise stable clinician impression.

03 FEB 2026
CBC/LFT safety review documented — Supports medication-safety monitoring without a documented abnormal lab signal.

15 FEB 2026
Highest recorded seizure burden in clinical-note window — 11 reported episodes make this the peak burden visit.

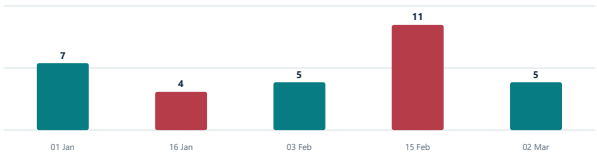
02 MAR 2026
Routine epilepsy follow-up with stable clinician impression — Contributes to longitudinal assessment of seizure burden, triggers, sleep, and tolerability.

10 SEP 2026
Pharmacogenomic (PGx) panel reported — 76 variants analyzed across 47 medications; see Pharmacogenomics.

03 Seizure Trends

Clinical-note episode counts, clustering, and duration

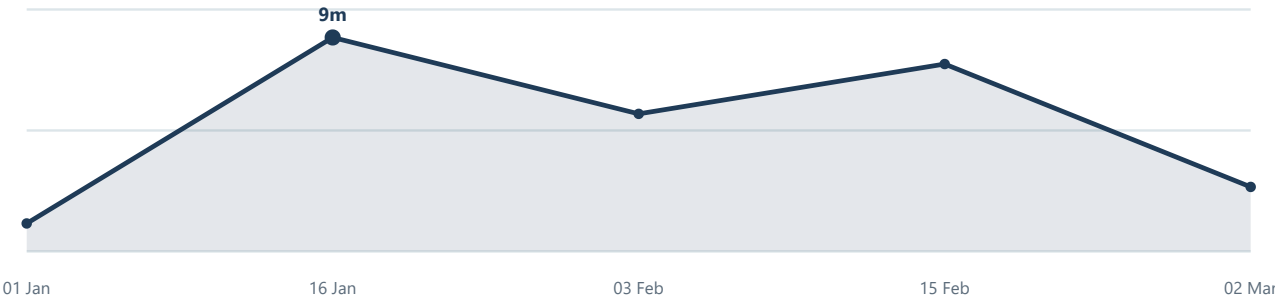
Reported Episodes per Visit



Red bars mark visits with a high-importance clinical event.

OVERALL PATTERN	Breakthrough seizure burden fluctuated across visits without a sustained monotonic worsening pattern.
HIGHEST BURDEN	11 episodes on 15 Feb 2026
LOWEST BURDEN	4 episodes on 16 Jan 2026
CLUSTERING	Yes
CLINICAL INTERPRETATION	Seizure risk remains clinically relevant because events persist on dual therapy and clustering was documented once.

Average Seizure Duration per Visit (minutes)



04 Aura & Triggers

Prodromal patterns and patient-reported trigger exposure

Persistent Aura / Prodromal Patterns

Deja vu frequency | Epigastric rising sensation | Feeling of derealization/detachment | Metallic taste in mouth | Racing heartbeat prior to episode | Sudden dizziness before seizure onset | Sudden intense anxiety or fear | Tingling or numbness sensations | Unusual smells (olfactory aura) | Visual distortions or flashing lights

Aura and prodromal symptoms remain useful warning signals for episode tracking.

Aura & Prodromal Symptoms — Questionnaire Domain

Latest score **71.7/100** (first 66.7, trend increasing)

- Did you experience an epigastric "rising" sensation in the stomach: **Frequent**
- Frequency of sudden, unprovoked psychological anxiety or panic feelings (ictal fear): **Rare**
- Frequency of sudden Deja vu or Jamais vu experiences: **Never**
- Did you notice an anomalous metallic, chemical, or bitter taste in your mouth: **Rare sensory distortion**
- Did you experience unprovoked olfactory distortions (strange, non-existent smells): **Never**

Epileptic Triggers — Lowest-Scoring Domain

Latest score **56.7/100** (min 56.7, trend stable)

- Rate your average psychological stress and workload level this month: **High/Distressing**
- Did you experience a delayed or completely missed dose of your anti-seizure medication: **1 missed dose**
- Exposure to flickering/flashing lights, high-contrast patterns, or prolonged screen glare: **Prolonged exposure**
- Daily average volume of dietary caffeine intake (coffee, tea, energy drinks): **1 cup**
- Any consumption of alcoholic beverages or exposure to hangover states this month: **Occasional drinking**

05 Medication Response

Current regimen, tolerability, and reported side effects

Current Regimen

DRUG	DOSE
Levetiracetam (Keppra)	1000 mg BID
Lamotrigine	150 mg BID

EFFECTIVENESS	Partial response
TOLERABILITY	Generally tolerated with intermittent dizziness and variable side-effect burden
DOCUMENTED TOXICITY	No
CLINICAL INTERPRETATION	Current antiseizure therapy is associated with stable clinician impressions but incomplete seizure suppression.

Persistent Side Effects (clinical notes)

Memory fog post-dose | Tremor in hands | Weight change since last visit | Gum hypertrophy | Mood irritability | Nausea after taking medication | Skin rash or itching | Daytime drowsiness | Dizziness/lightheadedness | Unsteady gait (ataxia)

Medication Side Effects — Questionnaire Domain

Latest score **78.3/100** (first 68.3, trend increasing)

- Rate the severity of daytime drowsiness or somnolence: **Mild (Manageable)**
- Experience of dizziness, vertigo, or lightheadedness after taking medication: **Never**
- Presence of physical instability, unsteadiness, or clumsy movements (ataxia): **Slight imbalance**
- Frequency of unprovoked irritability, anger, or mood volatility: **Mild irritability**
- Signs of dermatological distress or unexplained skin rashes: **Clear skin**

06 Drug Interactions

2 current medication(s) reconciled · Flockhart Cytochrome P450 Drug Interaction Table (curated-snapshot-2024)

DDI & Patient-Specific Therapy Assessment — Summary

Overall: 0 clinically significant | 0 resolved — no concern | 3 unresolved.

3	0	0	3
Total Findings Assessed	Clinically Significant	Resolved — No Concern	Unresolved

MEDICATION	FLAG	KEY FINDING	PATIENT IMPACT	EVIDENCE SOURCE
Lamotrigine	UNRESOLVED	Therapeutic evidence present, but clinical effectiveness remains incomplete.	Current antiseizure therapy is associated with stable clinician impressions but incomplete seizure suppression.	Patient pharmacogenomic findings; Laboratory findings; Clinical record; EEG findings
Levetiracetam (Keppra)	UNRESOLVED	Therapeutic evidence present, but clinical effectiveness remains incomplete.	Current antiseizure therapy is associated with stable clinician impressions but incomplete seizure suppression.	Patient pharmacogenomic findings; Laboratory findings; Clinical record; EEG findings

Key clinical interpretation: These flags identify findings that require ongoing clinical monitoring, not an automatic medication change. Each finding should be interpreted alongside the patient's actual treatment response, adverse effects, and the clinical/laboratory findings documented elsewhere in this report -- a resolved or unresolved flag is not itself a prescribing instruction.

Monitoring / Clinician Review Checklist

- Medication reconciliation (dose, frequency, and route consistent across all source records)
- Actual treatment response (seizure/symptom control on the current regimen)
- Relevant adverse effects reported since the last review
- Therapeutic drug levels, when indicated and available
- Relevant laboratory and clinical parameters (e.g. renal, hepatic, hematological)

Medication Reconciliation

DRUG	DOSE	FLAGS
Lamotrigine	150.0 mg BID	—
Levetiracetam (Keppra)	1000.0 mg BID	—

No conflicting doses found for the same drug across the reconciled source data.

Drug Interaction Flag Sheet — Extracted Drug List

Positive Impact = the pair's own evidence explicitly describes a favorable or well-tolerated combination, or explicitly states no clinically meaningful interaction; **Negative Impact** = the pair's own evidence documents a pharmacokinetic mechanism, a pharmacodynamic risk, or another stated risk. Patient PGx findings are shown as context only and never decide either column. A pair with no evidence, or whose evidence never clearly resolved either way, is not listed here at all -- it is never guessed into either column (see Known Limitations).

How pairs were classified: finding_agent (openai/gpt-5.5): the agent decides each pair's Impact and writes its Finding from that pair's own evidence (SKILL.md Part 6).

Counting: unique unordered drug pairs: A+B and B+A are one pair, counted once (the per-drug sections of the rendered report may show a pair under both of its drugs for readability; the totals never double-count it).

Where the PGx context comes from: patient-specific: this patient's own PGx panel (genotype cross-referenced against the PGx source report's gene-drug annotations) (Pharmacogenomic (PGx) Clinical Interpretation Report, 2026-09-10; 76 loci, 47 drugs). Pooled with external sources: no. Each finding keeps the PMID/PMCID citation the PGx report attaches to it, shown in its context line. CPIC and ClinVar are not pooled into this section (see the Pharmacogenomics section). context only -- PGx never decides a pair's Positive or Negative Impact (SKILL.md Part 3).

Sources excluded by the evidence checks: dailymed a - agent no finding: 1; dailymed a - agent unusable - output dropped hedge: 2; dailymed b - agent no finding: 1; dailymed b - agent unusable - output dropped hedge: 3; pubmed - agent no finding: 2; pubmed - agent unusable - output too long: 1; pubmed - animal or cell study: 9; pubmed - no usable abstract: 14; pubmed - not studied together: 84; pubmed - third drug: 12; tavily - agent no finding: 7; tavily - agent unusable - output too long: 1; tavily - animal or cell study: 1; tavily - no combination text: 83; tavily - page title or navigation only: 6; tavily - page unreadable: 184; tavily - third drug: 6; trials - no posted results: 213.

Screened 25 drug(s) across 325 unique pairs; 89 unique pair(s) resolved clearly to Positive or Negative and are shown below. 236 pair(s) had no evidence, or evidence that did not clearly resolve either way, and are not listed here -- never shown as a negative/no-concern finding (see Known Limitations).

ARIPIPRAZOLE

POSITIVE IMPACT	NEGATIVE IMPACT
Aripiprazole + Escitalopram **Findings:** Aripiprazole's label states that no dosage adjustment is necessary for CYP2C19 substrates, including escitalopram, when coadministered with aripiprazole. **Mechanism:** Escitalopram is listed as a CYP2C19 substrate in the aripiprazole label. **Direction:** No clinically important interaction requiring dose adjustment is stated. **Consequence:** No dosage adjustment is necessary for escitalopram when coadministered with aripiprazole. **Evidence:** DailyMed label (Drug Interactions section) [curated]; External evidence (PubMed) [inferred] **Hyperlink:** DailyMed	**Aripiprazole + Carbamazepine** **Findings:** Concomitant use of aripiprazole and carbamazepine decreased aripiprazole exposure compared with aripiprazole alone. The label recommends increasing the aripiprazole dosage when it is administered with a strong CYP3A4 inducer such as carbamazepine. **Mechanism:** carbamazepine is a strong CYP3A4 inducer affecting aripiprazole exposure **Direction:** decreased aripiprazole exposure **Consequence:** decreased exposure of aripiprazole compared to aripiprazole alone **Evidence:** DailyMed label (Drug Interactions section) [curated]; General web search (Tavily) [inferred]

POSITIVE IMPACT	NEGATIVE IMPACT

Aripiprazole + Lamotrigine

Findings: Aripiprazole's label states that no dosage adjustment is necessary for lamotrigine when coadministered with aripiprazole, based on pharmacokinetic interaction information.
Direction: no clinically important interaction/no dosage adjustment necessary
Consequence: no dosage adjustment is necessary for lamotrigine when coadministered with aripiprazole
Evidence: DailyMed label (Drug Interactions section) [curated]
Hyperlink: DailyMed

Aripiprazole + Lithium

Findings: Based on pharmacokinetic studies in the aripiprazole label, no dosage adjustment of aripiprazole is required when administered concomitantly with lithium, and no dosage adjustment is necessary for lithium when coadministered with aripiprazole.
Direction: no clinically important interaction requiring dosage adjustment
Consequence: no dosage adjustment is required for aripiprazole or lithium when coadministered
Evidence: DailyMed label (Drug Interactions section) [curated]
Hyperlink: DailyMed

Aripiprazole + Sertraline

Findings: The aripiprazole label states that no dosage adjustment is necessary for sertraline when sertraline is coadministered with aripiprazole.
Direction: no dosage adjustment necessary for sertraline
Evidence: DailyMed label (Drug Interactions section) [curated]
Hyperlink: DailyMed

Aripiprazole + Venlafaxine

Findings: Aripiprazole's label states that no dosage adjustment is necessary for CYP2D6 substrates, including venlafaxine, when coadministered with aripiprazole.
Mechanism: Venlafaxine is identified as a CYP2D6 substrate in the label's no clinically important interactions section.
Direction: No clinically important interaction requiring dosage adjustment is stated.
Consequence: No dosage adjustment is necessary when venlafaxine is coadministered with aripiprazole.
Evidence: DailyMed label (Drug Interactions section) [curated]; General web search (Tavily) [inferred]
Hyperlink: DailyMed

Hyperlink: DailyMed

Aripiprazole + Fluoxetine

Findings: Fluoxetine is listed as a strong CYP2D6 inhibitor that increases aripiprazole exposure when used concomitantly with aripiprazole. The label recommends reducing the aripiprazole dosage when aripiprazole is administered with a strong CYP2D6 inhibitor.
Mechanism: Fluoxetine is a strong CYP2D6 inhibitor; aripiprazole exposure is increased with strong CYP2D6 inhibitors.
Direction: Increased aripiprazole exposure compared with aripiprazole alone.
Consequence: Increased exposure of aripiprazole.
Evidence: DailyMed label (Drug Interactions section) [curated]; General web search (Tavily) [inferred]
Hyperlink: DailyMed

Aripiprazole + Oxcarbazepine

Findings: A case report stated that aripiprazole metabolism by CYP2D6 and CYP3A4 can be influenced by concomitant enzyme inducers or inhibitors, and described aripiprazole with oxcarbazepine, a known CYP3A4 inducer, as resulting in subtherapeutic aripiprazole levels and recurrence of manic symptoms in a patient with bipolar I disorder.
Mechanism: Aripiprazole metabolism by CYP2D6 and CYP3A4 can be influenced by concomitant medications that induce or inhibit these enzymes; oxcarbazepine is described as a known inducer of CYP3A4.
Direction: Subtherapeutic levels of aripiprazole were described with concomitant oxcarbazepine.
Consequence: Recurrence of manic symptoms in a patient with bipolar I disorder.
Evidence: General web search (Tavily) [inferred]
Hyperlink: General web search (Tavily)

Aripiprazole + Paroxetine

Findings: Paroxetine, listed as a strong CYP2D6 inhibitor, increases aripiprazole exposure when used concomitantly with aripiprazole compared with aripiprazole alone. The label recommends reducing the aripiprazole dosage when aripiprazole is administered with a strong CYP2D6 inhibitor.
Mechanism: Paroxetine is listed as a strong CYP2D6 inhibitor; concomitant use of aripiprazole with strong CYP2D6 inhibitors increased aripiprazole exposure.
Direction: Increased exposure of aripiprazole compared with aripiprazole alone.
Consequence: Increased aripiprazole exposure.
Evidence: DailyMed label (Drug Interactions section) [curated]; General web search (Tavily) [inferred]
Hyperlink: DailyMed

ATOMOXETINE

POSITIVE IMPACT	NEGATIVE IMPACT

No pairs currently fall in this category.

Atomoxetine + Paroxetine

Findings: Paroxetine may increase atomoxetine exposure because paroxetine is a CYP2D6 inhibitor and atomoxetine is listed as a CYP2D6 substrate. The label recommends decreasing the dosage of a CYP2D6 substrate if needed during concomitant paroxetine use, and increasing it if needed when paroxetine is discontinued.
Mechanism: Paroxetine is a CYP2D6 inhibitor; atomoxetine is listed as a CYP2D6 substrate.
Direction: Paroxetine may increase the exposure of atomoxetine.
Consequence: Increased exposure of the CYP2D6 substrate.
Evidence: DailyMed label (Drug Interactions section) [curated]; External evidence (PubMed) [inferred]; General web search (Tavily) [inferred]
Patient PGx context [predicted] (not part of this verdict): both drugs carry a finding for the shared gene HTR7. Atomoxetine HTR7 (GG): efficacy (PMID:30874608); Paroxetine HTR7 (GG): efficacy (PMID:30874608)
Hyperlink: DailyMed

Atomoxetine + Sertraline

Findings: Sertraline hydrochloride may increase atomoxetine exposure because sertraline is a CYP2D6 inhibitor and atomoxetine is listed as an example CYP2D6 substrate. The label recommends decreasing the dosage of a CYP2D6 substrate if needed during concomitant sertraline use, and increasing the substrate dosage may be needed if sertraline is discontinued.
Mechanism: Sertraline hydrochloride is a CYP2D6 inhibitor; atomoxetine is listed among CYP2D6 substrates.
Direction: Sertraline hydrochloride may increase the exposure of the CYP2D6 substrate atomoxetine.

POSITIVE IMPACT	NEGATIVE IMPACT
	Consequence: Increased exposure of atomoxetine may occur with concomitant sertraline hydrochloride use. **Evidence:** DailyMed label (Drug Interactions section) [curated] **Hyperlink:** DailyMed

ATORVASTATIN

POSITIVE IMPACT	NEGATIVE IMPACT
Atorvastatin + Nortriptyline **Findings:** In adults with migraine, atorvastatin added to nortriptyline for 24 weeks significantly reduced migraine attack frequency versus nortriptyline with placebo (P = 0.004) and improved quality of life at follow-up, with constipation the most common adverse effect in the atorvastatin arm. The study reported no effect on headache intensity. **Direction:** reduced migraine attack frequency and improved quality of life **Consequence:** potential prophylactic benefit for migraine headaches **Evidence:** External evidence (PubMed) [inferred] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene ABCB1. Atorvastatin ABCB1 (GG): toxicity (PMCID:PMC10587173); Nortriptyline ABCB1 (GG): toxicity (PMID:12082591) **Hyperlink:** PubMed	No pairs currently fall in this category.

BUPROPION

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	**Bupropion + Carbamazepine** **Findings:** Carbamazepine may induce the metabolism of bupropion and may decrease bupropion exposure; the bupropion label notes this was not systematically studied. If bupropion is used concomitantly with a CYP inducer such as carbamazepine, it may be necessary to increase the bupropion dose based on clinical response, without exceeding the maximum recommended dose. **Mechanism:** Carbamazepine may induce the metabolism of bupropion. **Direction:** May decrease bupropion exposure. **Evidence:** DailyMed label (Drug Interactions section) [curated]; General web search (Tavily) [inferred] **Hyperlink:** DailyMed **Bupropion + Fluoxetine** **Findings:** Bupropion inhibits CYP2D6 and can increase exposure of CYP2D6 substrates, including fluoxetine; the label states that dose reduction of these CYP2D6 substrates may be necessary when used concomitantly with bupropion, particularly for drugs with a narrow therapeutic index. **Mechanism:** Bupropion and its metabolites are CYP2D6 inhibitors; fluoxetine is listed among drugs metabolized by CYP2D6. **Direction:** Can increase concentrations/exposures of CYP2D6 substrates, including fluoxetine. **Evidence:** DailyMed label (Drug Interactions section) [curated] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene HTR2A. Bupropion HTR2A (TT): reduced efficacy (PMID:22947179); Fluoxetine HTR2A (AA): reduced efficacy (PMID:32073889); Fluoxetine HTR2A (TT): toxicity (PMID:16710319) **Hyperlink:** DailyMed **Bupropion + Haloperidol** **Findings:** Bupropion inhibits CYP2D6 and can increase exposure of CYP2D6 substrates, including the antipsychotic haloperidol. The label recommends considering dose reduction of CYP2D6 substrates when used concomitantly with bupropion, particularly for drugs with a narrow therapeutic index. **Mechanism:** Bupropion and its metabolites are CYP2D6 inhibitors; haloperidol is listed among antipsychotics that are CYP2D6 substrates. **Direction:** Bupropion can increase concentrations/exposures of haloperidol. **Consequence:** Increased exposure of haloperidol as a CYP2D6 substrate. **Evidence:** DailyMed label (Drug Interactions section) [curated]; External evidence (PubMed) [inferred]; General web search (Tavily) [inferred] **Hyperlink:** DailyMed **Bupropion + Nortriptyline** **Findings:** Bupropion inhibits CYP2D6 and can increase concentrations/exposures of CYP2D6 substrates, including nortriptyline. The label recommends considering dose reduction or decreasing the dose of these CYP2D6 substrates when used concomitantly with bupropion, particularly for drugs with a narrow therapeutic index.

POSITIVE IMPACT	NEGATIVE IMPACT
	Mechanism: Bupropion and its metabolites are CYP2D6 inhibitors; nortriptyline is listed among drugs metabolized by CYP2D6/CYP2D6 substrates. **Direction:** Bupropion can increase nortriptyline concentrations/exposures. **Consequence:** Increased exposure of nortriptyline as a CYP2D6 substrate when coadministered with bupropion. **Evidence:** DailyMed label (Drug Interactions section) [curated]; General web search (Tavily) [inferred] **Hyperlink:** DailyMed
	Bupropion + Paroxetine **Findings:** Bupropion inhibits CYP2D6 and can increase exposure to CYP2D6 substrates, including paroxetine. The label recommends considering dose reduction of CYP2D6 substrates when used concomitantly with bupropion, particularly for drugs with a narrow therapeutic index. **Mechanism:** Bupropion and its metabolites are CYP2D6 inhibitors; paroxetine is listed among antidepressants metabolized by CYP2D6. **Direction:** Bupropion can increase paroxetine exposure. **Consequence:** Increased exposure of paroxetine as a CYP2D6 substrate. **Evidence:** DailyMed label (Drug Interactions section) [curated] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene HTR2A. Bupropion HTR2A (TT): reduced efficacy (PMID:22947179); Paroxetine HTR2A (TT): toxicity (PMID:16710319) **Hyperlink:** DailyMed
	Bupropion + Phenytoin **Findings:** Phenytoin is listed as a CYP2B6 inducer that may induce bupropion metabolism and may decrease bupropion exposure, although the label notes this was not systematically studied. If bupropion is used concomitantly with a CYP inducer such as phenytoin, the bupropion dose may need to be increased based on clinical response, without exceeding the maximum recommended dose. **Mechanism:** Phenytoin is a CYP2B6/CYP inducer and may induce the metabolism of bupropion. **Direction:** May decrease bupropion exposure. **Consequence:** Reduced bupropion exposure may require action based on clinical response. **Evidence:** DailyMed label (Drug Interactions section) [curated] **Hyperlink:** DailyMed
	Bupropion + Sertraline **Findings:** Bupropion inhibits CYP2D6 and can increase exposure to CYP2D6 substrates, including sertraline. The label says dose reduction of these CYP2D6 substrates may be necessary when used concomitantly with bupropion, particularly for drugs with a narrow therapeutic index. **Mechanism:** Bupropion and its metabolites are CYP2D6 inhibitors; sertraline is listed among antidepressants that are CYP2D6 substrates. **Direction:** Bupropion can increase concentrations/exposures of CYP2D6 substrates, including sertraline. **Consequence:** Increased exposure to sertraline as a CYP2D6 substrate. **Evidence:** DailyMed label (Drug Interactions section) [curated] **Hyperlink:** DailyMed
	Bupropion + Venlafaxine **Findings:** Bupropion inhibits CYP2D6 and can increase exposure of CYP2D6 substrates, including venlafaxine; the label says dose reduction of these CYP2D6 substrates may be necessary when used concomitantly with bupropion, particularly for drugs with a narrow therapeutic index. **Mechanism:** Bupropion and its metabolites are CYP2D6 inhibitors; venlafaxine is listed among drugs metabolized by CYP2D6. **Direction:** Bupropion can increase concentrations/exposures of venlafaxine as a CYP2D6 substrate. **Evidence:** DailyMed label (Drug Interactions section) [curated] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene HTR2A. Bupropion HTR2A (TT): reduced efficacy (PMID:22947179); Venlafaxine HTR2A (AG): efficacy (PMID:22006095) **Hyperlink:** DailyMed

CARBAMAZEPINE

POSITIVE IMPACT	NEGATIVE IMPACT
Carbamazepine + Levetiracetam **Findings:** Levetiracetam did not influence carbamazepine plasma concentrations, and carbamazepine did not influence levetiracetam pharmacokinetics in placebo-controlled clinical studies evaluating levetiracetam with other antiepileptic drugs.	**Carbamazepine + Escitalopram** **Findings:** Carbamazepine did not significantly affect carbamazepine pharmacokinetics when administered with racemic citalopram, but the escitalopram label says carbamazepine might increase escitalopram clearance if carbamazepine

POSITIVE IMPACT	NEGATIVE IMPACT
Direction: no effect on plasma concentration/pharmacokinetics **Consequence:** no clinically meaningful pharmacokinetic interaction is indicated by the label data **Evidence:** DailyMed label (Drug Interactions section) [curated]; General web search (Tavily) [inferred] **Hyperlink:** DailyMed	and escitalopram are coadministered. The label recommends considering this possibility because of carbamazepine's enzyme-inducing properties. **Mechanism:** Carbamazepine's enzyme-inducing properties may increase escitalopram clearance. **Direction:** Carbamazepine might increase the clearance of escitalopram. **Consequence:** Potential decreased escitalopram exposure/clearance increase if coadministered. **Evidence:** DailyMed label (Drug Interactions section) [curated] **Hyperlink:** DailyMed
	Carbamazepine + Fluoxetine **Findings:** General web evidence states that co-administered carbamazepine accelerates hepatic metabolism of fluoxetine and norfluoxetine, reducing fluoxetine plasma concentrations by 50% or more over 2–4 weeks. It identifies potential loss of antidepressant efficacy and recommends close monitoring of psychiatric symptoms and considering a 25–50% fluoxetine dose increase with careful titration. **Mechanism:** Carbamazepine accelerates hepatic metabolism of fluoxetine and norfluoxetine via enzyme induction. **Direction:** Fluoxetine plasma concentrations are significantly reduced, by 50% or more. **Consequence:** Potential loss of antidepressant efficacy, return of depressive symptoms, increased anxiety, or worsening psychiatric condition; fluoxetine levels may increase with serotonergic side effects if carbamazepine is discontinued. **Evidence:** DailyMed label (Drug Interactions section) [curated]; General web search (Tavily) [inferred] **Hyperlink:** General web search (Tavily)
	Carbamazepine + Haloperidol **Findings:** Coadministration of haloperidol with potent CYP3A4 enzyme inducers, including carbamazepine, may gradually decrease haloperidol plasma concentrations to an extent that efficacy may be reduced. **Mechanism:** Carbamazepine is listed as a potent enzyme inducer of CYP3A4; haloperidol exposure may be decreased when coadministered with such inducers. **Direction:** May gradually decrease haloperidol plasma concentrations. **Consequence:** Haloperidol efficacy may be reduced. **Evidence:** DailyMed label (Drug Interactions section) [curated] **Hyperlink:** DailyMed
	Carbamazepine + Lamotrigine **Findings:** Carbamazepine decreases lamotrigine concentrations by approximately 40% by inducing lamotrigine glucuronidation; the label also states carbamazepine may increase carbamazepine epoxide levels. **Mechanism:** Carbamazepine induces lamotrigine glucuronidation. **Direction:** Decreased lamotrigine concentrations by approximately 40%; may increase carbamazepine epoxide levels. **Consequence:** Decreased lamotrigine exposure/concentrations; possible increased carbamazepine epoxide levels. **Evidence:** DailyMed label (Drug Interactions section) [curated] **Hyperlink:** DailyMed
	Carbamazepine + Nortriptyline **Findings:** Concomitant carbamazepine 500–600 mg/day decreased serum nortriptyline concentrations from a prior mean of 355 ± 49 nM to 140 and 134 nM; after the nortriptyline dose was increased to 150 mg/day, concentrations rose and the depression disappeared. The case report concludes that carbamazepine induces nortriptyline metabolism. **Mechanism:** Carbamazepine induces nortriptyline metabolism. **Direction:** Serum nortriptyline concentrations decreased during concomitant carbamazepine intake. **Consequence:** The patient's depression persisted while serum nortriptyline concentrations were decreased during concomitant carbamazepine intake. **Evidence:** External evidence (PubMed) [inferred]; General web search (Tavily) [inferred] **Hyperlink:** PubMed
	Carbamazepine + Olanzapine **Findings:** Carbamazepine therapy (200 mg twice daily) causes an approximately 50% increase in olanzapine clearance, likely because carbamazepine is a potent inducer of CYP1A2 activity. Higher daily doses of carbamazepine may cause an even greater increase in olanzapine clearance. **Mechanism:** Carbamazepine is a potent inducer of CYP1A2 activity. **Direction:** Approximately 50% increase in olanzapine clearance with carbamazepine 200 mg twice daily; higher daily doses of carbamazepine may cause an even greater increase.

POSITIVE IMPACT	NEGATIVE IMPACT

Consequence: Increased clearance of olanzapine.
Evidence: DailyMed label (Drug Interactions section) [curated]
Hyperlink: DailyMed

Carbamazepine + Oxcarbazepine

Findings: Carbamazepine administered concurrently with oxcarbazepine (Oxtellar XR) decreased blood levels of oxcarbazepine's active metabolite MHD. The label recommends monitoring MHD plasma levels during Oxtellar XR titration and says an Oxtellar XR dosage adjustment may be required after starting, changing, or stopping carbamazepine.
Mechanism: Carbamazepine is listed as a strong CYP3A4 inducer or UGT inducer affecting Oxtellar XR.
Direction: Decreased blood levels of oxcarbazepine's active metabolite MHD.
Evidence: DailyMed label (Drug Interactions section) [curated]; External evidence (PubMed) [inferred]
Hyperlink: DailyMed

Carbamazepine + Perampanel

Findings: Carbamazepine, as a moderate or strong CYP3A4 inducer, decreased perampanel plasma levels by approximately 50% to 67% when used concomitantly. The label recommends increasing perampanel starting doses in the presence of such inducers and closely monitoring clinical response and tolerability when they are introduced or withdrawn, because perampanel dose adjustment may be necessary.
Mechanism: Carbamazepine is included among moderate and strong CYP3A4 inducers that increase clearance of perampanel.
Direction: Decreased perampanel plasma levels by approximately 50% to 67%.
Consequence: Reduced perampanel plasma concentrations; clinical response and tolerability should be monitored.
Evidence: DailyMed label (Drug Interactions section) [curated]; External evidence (PubMed) [inferred]
Hyperlink: DailyMed

Carbamazepine + Phenytoin

Findings: Carbamazepine is listed in phenytoin's label among antiepileptic drugs that may decrease phenytoin serum levels.
Direction: may decrease phenytoin serum levels
Consequence: decreased phenytoin serum levels
Evidence: DailyMed label (Drug Interactions section) [curated]; General web search (Tavily) [inferred]
Patient PGx context [predicted] (not part of this verdict): both drugs carry a finding for the shared gene CYP1A1, GABRA1. Carbamazepine CYP1A1 (CA): toxicity (PMID:21121773); Phenytoin CYP1A1 (CA): toxicity (PMID:21121773); Carbamazepine GABRA1 (GA): reduced efficacy (PMCID:PMC3125053); Phenytoin GABRA1 (GA): reduced efficacy (PMCID:PMC3125053)
Hyperlink: DailyMed

Carbamazepine + Quetiapine

Findings: Carbamazepine is listed as a prototype CYP3A4 inducer that decreases quetiapine exposure, and the quetiapine label states dose adjustment will be necessary if quetiapine is co-administered with potent CYP3A4 inducers. For CYP3A4 induction, the label notes that coadministration with phenytoin increased quetiapine oral clearance 5-fold and that increased quetiapine doses up to 5-fold may be needed.
Mechanism: Carbamazepine is a prototype CYP3A4 inducer; quetiapine exposure is decreased by prototype CYP3A4 inducers.
Direction: Decreased quetiapine exposure.
Evidence: DailyMed label (Drug Interactions section) [curated]; External evidence (PubMed) [inferred]; General web search (Tavily) [inferred]
Hyperlink: DailyMed

Carbamazepine + Sertraline

Findings: General web evidence states that carbamazepine coadministered with sertraline accelerates hepatic breakdown of sertraline, reducing sertraline plasma concentrations. It states this may reduce sertraline effectiveness, with possible worsening psychiatric symptoms, and recommends close symptom monitoring and possible sertraline dose increase.
Mechanism: Carbamazepine enzyme induction accelerates hepatic breakdown of sertraline.
Direction: Reduced sertraline plasma concentrations.
Consequence: Reduced therapeutic effectiveness of sertraline with possible worsening depression, anxiety, or other psychiatric symptoms.
Evidence: General web search (Tavily) [inferred]
Hyperlink: General web search (Tavily)

POSITIVE IMPACT	NEGATIVE IMPACT
	Carbamazepine + Valproic Acid **Findings:** Carbamazepine can increase valproate clearance; the Valproic Acid label states that carbamazepine can double valproate clearance. The label recommends increased monitoring of valproate and concomitant drug concentrations and dosage adjustment when enzyme-inducing drugs such as carbamazepine are introduced or withdrawn. **Mechanism:** Carbamazepine is a hepatic enzyme-inducing drug that elevates levels of glucuronosyltransferases and increases valproate clearance. **Direction:** Carbamazepine can double the clearance of valproate. **Consequence:** Patients receiving polytherapy with antiepilepsy drugs have shorter valproate half-lives and lower concentrations than patients on monotherapy. **Evidence:** DailyMed label (Drug Interactions section) [curated] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene CYP1A1, GABRA1. Carbamazepine CYP1A1 (CA): toxicity (PMID:21121773); Valproic Acid CYP1A1 (CA): toxicity (PMID:21121773); Carbamazepine GABRA1 (GA): reduced efficacy (PMCID:PMC3125053); Valproic Acid GABRA1 (GA): reduced efficacy (PMCID:PMC3125053) **Hyperlink:** DailyMed

CITALOPRAM

POSITIVE IMPACT	NEGATIVE IMPACT
Citalopram + Methylphenidate **Findings:** In ClinicalTrials.gov trial NCT00794040, add-on citalopram following optimized methylphenidate produced a statistically significant improvement in irritability CGI-I response compared with add-on placebo following optimized methylphenidate (p=0.006). **Direction:** statistically significant improvement in irritability CGI-I response **Evidence:** ClinicalTrials.gov [inferred] **Hyperlink:** ClinicalTrials.gov	**Citalopram + Escitalopram** **Findings:** Citalopram and escitalopram have additive serotonergic activity when combined, increasing the risk of serotonin syndrome, activation, sleep disturbance, and agitation. **Mechanism:** Additive serotonergic activity from combined serotonergic agents. **Direction:** Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. **Consequence:** Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. **Evidence:** Pharmacodynamic interaction reference [curated]; DailyMed label (Drug Interactions section) [curated] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene HTR1B, HTR2A. Citalopram HTR1B (CG): toxicity (PMID:27324805); Escitalopram HTR1B (AC): toxicity (PMCID:PMC3967864); Citalopram HTR2A (TT): toxicity (PMID:16710319); Escitalopram HTR2A (TT): toxicity (PMCID:PMC3967864) **Citalopram + Fluoxetine** **Findings:** Citalopram and fluoxetine have additive serotonergic activity when combined, increasing the risk of serotonin syndrome, activation, sleep disturbance, and agitation. **Mechanism:** Additive serotonergic activity from combined serotonergic agents. **Direction:** Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. **Consequence:** Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. **Evidence:** Pharmacodynamic interaction reference [curated] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene GSK3B, HTR2A, SERPINE1, SRP19. Citalopram GSK3B (AG): reduced efficacy (PMID:18195729); Fluoxetine GSK3B (AG): reduced efficacy (PMID:18195729); Citalopram HTR2A (TT): toxicity (PMID:16710319); Fluoxetine HTR2A (AA): reduced efficacy (PMID:32073889); Fluoxetine HTR2A (TT): toxicity (PMID:16710319); Citalopram SERPINE1 (GG): reduced efficacy (PMID:18794724); Fluoxetine SERPINE1 (GG): reduced efficacy (PMID:18794724); Citalopram SRP19 (GG): efficacy (PMID:22795047); Fluoxetine SRP19 (GG): efficacy (PMID:22795047) **Citalopram + Lithium** **Findings:** Concomitant use of citalopram with lithium, an other serotonergic drug listed in the label, increases the risk of serotonin syndrome. The label recommends monitoring for signs and symptoms of serotonin syndrome, particularly during citalopram initiation and dosage increases, and considering discontinuation of citalopram and/or concomitant serotonergic drugs if serotonin syndrome occurs. **Mechanism:** Both citalopram and lithium are serotonergic drugs. **Direction:** increases the risk of serotonin syndrome **Consequence:** serotonin syndrome **Evidence:** DailyMed label (Drug Interactions section) [curated] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene FKBP5, GSK3B, HTR1B. Citalopram FKBP5 (AG): efficacy (PMCID:PMC2587308); Lithium FKBP5 (TC): toxicity (PMID:21449676); Citalopram GSK3B (AG): reduced efficacy (PMID:18195729); Lithium GSK3B (AG): efficacy (PMID:23021822); Citalopram HTR1B (CG): toxicity (PMID:27324805); Lithium HTR1B (TA): efficacy (PMID:21449676)

POSITIVE IMPACT	NEGATIVE IMPACT
	Hyperlink: DailyMed
	Citalopram + Paroxetine **Findings:** Citalopram and paroxetine have additive serotonergic activity when combined, increasing the risk of serotonin syndrome, activation, sleep disturbance, and agitation. **Mechanism:** Additive serotonergic activity from combined serotonergic agents. **Direction:** Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. **Consequence:** Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. **Evidence:** Pharmacodynamic interaction reference [curated]; General web search (Tavily) [inferred] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene HTR1B, HTR2A, SRP19. Citalopram HTR1B (CG): toxicity (PMID:27324805); Paroxetine HTR1B (TA): toxicity (PMID:21449676); Citalopram HTR2A (TT): toxicity (PMID:16710319); Paroxetine HTR2A (TT): toxicity (PMID:16710319); Citalopram SRP19 (GG): efficacy (PMID:22795047); Paroxetine SRP19 (GG): efficacy (PMID:22795047)
	Citalopram + Sertraline **Findings:** Citalopram and sertraline have additive serotonergic activity when combined, increasing the risk of serotonin syndrome, activation, sleep disturbance, and agitation. **Mechanism:** Additive serotonergic activity from combined serotonergic agents. **Direction:** Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. **Consequence:** Increased risk of serotonin syndrome, activation, sleep disturbance, and agitation. **Evidence:** Pharmacodynamic interaction reference [curated]; General web search (Tavily) [inferred] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene SRP19. Citalopram SRP19 (GG): efficacy (PMID:22795047); Sertraline SRP19 (GG): efficacy (PMID:22795047)

DULOXETINE

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	**Duloxetine + Fluoxetine** **Findings:** Duloxetine exposure is expected to increase when used with fluoxetine, because duloxetine's label states that paroxetine increased duloxetine AUC by about 60% and similar effects would be expected with other potent CYP2D6 inhibitors such as fluoxetine. **Direction:** Duloxetine AUC increased by about 60% with paroxetine, and similar effects would be expected with fluoxetine. **Consequence:** Increased duloxetine concentration/exposure is expected with potent CYP2D6 inhibitors such as fluoxetine. **Evidence:** DailyMed label (Drug Interactions section) [curated] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene HTR7. Duloxetine HTR7 (GG): efficacy (PMID:30874608); Fluoxetine HTR7 (GG): efficacy (PMID:30874608) **Hyperlink:** DailyMed
	Duloxetine + Paroxetine **Findings:** Potent CYP2D6 inhibitors may increase duloxetine concentrations; in the label, concomitant use of duloxetine 40 mg once daily with paroxetine 20 mg once daily increased duloxetine AUC by about 60%, with greater degrees of inhibition expected at higher paroxetine doses. **Mechanism:** Paroxetine is a potent CYP2D6 inhibitor; duloxetine exposure is affected by CYP2D6 inhibition. **Direction:** Duloxetine AUC increased by about 60% with paroxetine 20 mg once daily, with greater inhibition expected at higher paroxetine doses. **Consequence:** May increase duloxetine concentrations. **Evidence:** DailyMed label (Drug Interactions section) [curated] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene HTR7. Duloxetine HTR7 (GG): efficacy (PMID:30874608); Paroxetine HTR7 (GG): efficacy (PMID:30874608) **Hyperlink:** DailyMed
	Duloxetine + Sertraline **Findings:** General web evidence says duloxetine and sertraline combination therapy requires close monitoring for signs and symptoms of serotonin syndrome, especially during initiation, dose changes, or addition of other serotonergic medications. It recommends patient education, considering lower starting doses with slow titration if

POSITIVE IMPACT	NEGATIVE IMPACT
	combination therapy is necessary, and frequent monitoring or alternative treatment options. **Direction:** requires close monitoring for signs and symptoms of serotonin syndrome **Consequence:** serotonin syndrome **Evidence:** General web search (Tavily) [inferred] **Hyperlink:** General web search (Tavily)

ESCITALOPRAM

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	**Escitalopram + Fluoxetine** **Findings:** Escitalopram and fluoxetine have additive serotonergic activity when combined, increasing the risk of serotonin syndrome, activation, sleep disturbance, and agitation. **Mechanism:** Additive serotonergic activity from combined serotonergic agents. **Direction:** Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. **Consequence:** Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. **Evidence:** Pharmacodynamic interaction reference [curated] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene HTR2A. Escitalopram HTR2A (TT): toxicity (PMCID:PMC3967864); Fluoxetine HTR2A (AA): reduced efficacy (PMID:32073889); Fluoxetine HTR2A (TT): toxicity (PMID:16710319) **Escitalopram + Lithium** **Findings:** Concomitant use of escitalopram with lithium, listed by the label as another serotonergic drug, increases the risk of serotonin syndrome. The label recommends monitoring for signs and symptoms of serotonin syndrome, particularly during escitalopram initiation and dosage increases, and considering discontinuation of escitalopram and/or concomitant serotonergic drugs if serotonin syndrome occurs. **Mechanism:** escitalopram and lithium are serotonergic drugs **Direction:** increases the risk of serotonin syndrome **Consequence:** serotonin syndrome **Evidence:** DailyMed label (Drug Interactions section) [curated]; General web search (Tavily) [inferred] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene HTR1B. Escitalopram HTR1B (AC): toxicity (PMCID:PMC3967864); Lithium HTR1B (TA): efficacy (PMID:21449676) **Hyperlink:** DailyMed **Escitalopram + Paroxetine** **Findings:** Escitalopram and paroxetine have additive serotonergic activity when combined, increasing the risk of serotonin syndrome, activation, sleep disturbance, and agitation. **Mechanism:** Additive serotonergic activity from combined serotonergic agents **Direction:** Increased serotonin-syndrome risk, activation, sleep disturbance, agitation **Consequence:** Increased risk of serotonin syndrome, activation, sleep disturbance, and agitation **Evidence:** Pharmacodynamic interaction reference [curated] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene BDNF, HTR1B, HTR2A. Escitalopram BDNF (TC): toxicity (PMID:19641488); Paroxetine BDNF (CT): reduced efficacy (PMID:24577123); Escitalopram HTR1B (AC): toxicity (PMCID:PMC3967864); Paroxetine HTR1B (TA): toxicity (PMID:21449676); Escitalopram HTR2A (TT): toxicity (PMCID:PMC3967864); Paroxetine HTR2A (TT): toxicity (PMID:16710319) **Escitalopram + Sertraline** **Findings:** Escitalopram and sertraline have additive serotonergic activity when combined, increasing the risk of serotonin syndrome, activation, sleep disturbance, and agitation. **Mechanism:** Additive serotonergic activity from combined serotonergic agents. **Direction:** Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. **Consequence:** Increased risk of serotonin syndrome, activation, sleep disturbance, and agitation. **Evidence:** Pharmacodynamic interaction reference [curated]

ETHOSUXIMIDE

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	**Ethosuximide + Phenytoin**

POSITIVE IMPACT	NEGATIVE IMPACT
	Findings: Ethosuximide may elevate phenytoin serum levels when concurrently administered. The label recommends periodic serum level determinations of concurrently administered antiepileptic drugs because ethosuximide may interact with them. **Direction:** Ethosuximide may elevate phenytoin serum levels. **Evidence:** DailyMed label (Drug Interactions section) [curated]; DailyMed label (Drug Interactions section) [curated]; General web search (Tavily) [inferred] **Hyperlink:** DailyMed
	Ethosuximide + Valproic Acid **Findings:** Valproic acid has been reported to both increase and decrease ethosuximide levels. The ethosuximide label says periodic serum level determinations of concurrently administered antiepileptic drugs may be necessary. **Direction:** Valproic acid has been reported to both increase and decrease ethosuximide levels. **Evidence:** DailyMed label (Drug Interactions section) [curated]; DailyMed label (Drug Interactions section) [curated] **Hyperlink:** DailyMed

FLUOXETINE

POSITIVE IMPACT	NEGATIVE IMPACT
Fluoxetine + Olanzapine **Findings:** In ClinicalTrials.gov trial NCT00844857, olanzapine/fluoxetine combination significantly improved the primary CDRS-R total score at Week 8 compared with placebo in patients ages 10–17 with bipolar depression (p=0.003). **Direction:** improved CDRS-R total score at Week 8 compared with placebo **Evidence:** DailyMed label (Drug Interactions section) [curated]; DailyMed label (Drug Interactions section) [curated]; ClinicalTrials.gov [inferred]; General web search (Tavily) [inferred] **Hyperlink:** ClinicalTrials.gov	**Fluoxetine + Haloperidol** **Findings:** Fluoxetine is listed as a combined CYP3A4 and CYP2D6 inhibitor that, when coadministered with haloperidol, can increase haloperidol plasma concentrations. The label states that increased haloperidol concentrations may increase adverse events, including QTc interval prolongation, and recommends monitoring for increased or prolonged haloperidol effects and decreasing the haloperidol injection dose as needed. **Mechanism:** Fluoxetine is a combined CYP3A4 and CYP2D6 inhibitor; inhibition of haloperidol metabolism via these routes may increase haloperidol concentrations. **Direction:** Increased haloperidol plasma concentrations. **Consequence:** May increase adverse events, including QTc interval prolongation. **Evidence:** DailyMed label (Drug Interactions section) [curated]; DailyMed label (Drug Interactions section) [curated] **Hyperlink:** DailyMed
	Fluoxetine + Lithium **Findings:** Fluoxetine's label states that concomitant use of serotonergic drugs, including lithium, with fluoxetine increases the risk of serotonin syndrome. It recommends monitoring for signs and symptoms, particularly during treatment initiation and dosage increases, and considering discontinuation of fluoxetine and/or concomitant serotonergic drugs if serotonin syndrome occurs. **Mechanism:** Concomitant serotonergic drugs, including lithium, used with fluoxetine **Direction:** increases the risk **Consequence:** serotonin syndrome **Evidence:** DailyMed label (Drug Interactions section) [curated] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene GSK3B. Fluoxetine GSK3B (AG): reduced efficacy (PMID:18195729); Lithium GSK3B (AG): efficacy (PMID:23021822) **Hyperlink:** DailyMed
	Fluoxetine + Nortriptyline **Findings:** Fluoxetine may inhibit CYP2D6 and increase nortriptyline (a TCA) plasma exposure; the label notes TCA exposure increases may be small or as large as an 8-fold increase in AUC and that a stable TCA patient may become abruptly toxic when a CYP2D6 inhibitor is added. The label advises caution with coadministration, possible lower doses, and monitoring TCA plasma levels. **Mechanism:** Fluoxetine, an SSRI, inhibits cytochrome P450 2D6; nortriptyline is a tricyclic antidepressant affected by CYP2D6 inhibition. **Direction:** May increase nortriptyline/TCA plasma concentration or exposure, with increases described as small or up to an 8-fold increase in TCA AUC. **Consequence:** A patient stable on a TCA may become abruptly toxic when a CYP2D6-inhibiting drug is added as concomitant therapy. **Evidence:** DailyMed label (Drug Interactions section) [curated] **Hyperlink:** DailyMed
	Fluoxetine + Paroxetine **Findings:** Fluoxetine and paroxetine have additive serotonergic activity when combined, increasing the risk of serotonin syndrome, activation, sleep disturbance, and agitation. **Mechanism:** Additive serotonergic activity from combined serotonergic agents. **Direction:** Increased serotonin-syndrome risk, activation, sleep disturbance, agitation.

POSITIVE IMPACT	NEGATIVE IMPACT
	Consequence: Increased risk of serotonin syndrome, activation, sleep disturbance, and agitation. **Evidence:** Pharmacodynamic interaction reference [curated] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene HTR2A, HTR7, SRP19. Fluoxetine HTR2A (AA): reduced efficacy (PMID:32073889); Fluoxetine HTR2A (TT): toxicity (PMID:16710319); Paroxetine HTR2A (TT): toxicity (PMID:16710319); Fluoxetine HTR7 (GG): efficacy (PMID:30874608); Paroxetine HTR7 (GG): efficacy (PMID:30874608); Fluoxetine SRP19 (GG): efficacy (PMID:22795047); Paroxetine SRP19 (GG): efficacy (PMID:22795047)
	Fluoxetine + Phenytoin **Findings:** Fluoxetine may increase phenytoin serum levels; the phenytoin label states that adding or withdrawing such agents in patients on phenytoin therapy may require adjustment of the phenytoin dose to achieve optimal clinical outcome. **Direction:** Fluoxetine may increase phenytoin serum levels. **Evidence:** DailyMed label (Drug Interactions section) [curated]; DailyMed label (Drug Interactions section) [curated] **Hyperlink:** DailyMed
	Fluoxetine + Quetiapine **Findings:** General web evidence states that fluoxetine and quetiapine may interact through fluoxetine inhibition of CYP2D6, which can lead to increased quetiapine plasma concentrations and prolonged elimination. It also states that both drugs can independently prolong the QT interval, so combined use may have additive effects on cardiac conduction. **Mechanism:** Fluoxetine is a potent inhibitor of CYP2D6, which the source states is involved in quetiapine metabolism; both drugs can independently prolong the QT interval. **Direction:** Can lead to increased quetiapine plasma concentrations and prolonged elimination; combined use may have additive effects on cardiac conduction. **Consequence:** Potential additive QT-interval effects on cardiac conduction. **Evidence:** General web search (Tavily) [inferred] **Hyperlink:** General web search (Tavily)
	Fluoxetine + Sertraline **Findings:** Fluoxetine and sertraline have additive serotonergic activity when combined, increasing the risk of serotonin syndrome, activation, sleep disturbance, and agitation. **Mechanism:** Additive serotonergic activity from combined serotonergic agents. **Direction:** Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. **Consequence:** Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. **Evidence:** Pharmacodynamic interaction reference [curated]; General web search (Tavily) [inferred] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene SRP19. Fluoxetine SRP19 (GG): efficacy (PMID:22795047); Sertraline SRP19 (GG): efficacy (PMID:22795047)

HALOPERIDOL

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	**Haloperidol + Paroxetine** **Findings:** Paroxetine is listed as a CYP2D6 inhibitor that can increase haloperidol plasma concentrations when coadministered with haloperidol; increased haloperidol concentrations may increase adverse-event risk, including QTc interval prolongation. The label recommends monitoring for increased or prolonged haloperidol effects and decreasing the haloperidol injection dose as deemed necessary. **Mechanism:** Paroxetine is a CYP2D6 inhibitor; inhibition of haloperidol metabolism through CYP2D6 may increase haloperidol concentrations. **Direction:** Increased haloperidol plasma concentrations. **Consequence:** Increased risk of adverse events, including QTc interval prolongation. **Evidence:** DailyMed label (Drug Interactions section) [curated] **Hyperlink:** DailyMed
	Haloperidol + Phenytoin **Findings:** Coadministration of haloperidol with potent CYP3A4 enzyme inducers, including phenytoin, may gradually decrease haloperidol plasma concentrations to an extent that efficacy may be reduced. The haloperidol label recommends monitoring for signs or symptoms of reduced therapeutic effects when an inducer is added and adjusting the haloperidol dose as necessary. **Mechanism:** Phenytoin is included among potent enzyme inducers of CYP3A4 that may decrease haloperidol plasma concentrations. **Direction:** May gradually decrease haloperidol plasma concentrations.

POSITIVE IMPACT	NEGATIVE IMPACT
	Consequence: Haloperidol efficacy may be reduced. **Evidence:** DailyMed label (Drug Interactions section) [curated]; General web search (Tavily) [inferred] **Hyperlink:** DailyMed
	Haloperidol + Quetiapine **Findings:** General web evidence states that combining haloperidol and quetiapine has primary risks that include increased sedation and drowsiness, which may impair cognitive function and increase fall risk, especially in elderly patients. The same source states that both drugs can prolong the QT interval, potentially increasing the risk of serious cardiac arrhythmias, and that enhanced dopamine blockade increases the risk of extrapyramidal symptoms. **Mechanism:** Both drugs block dopamine D2 receptors; both medications can prolong the QT interval through blockade of cardiac potassium channels (hERG channels); quetiapine has antihistaminergic properties and haloperidol has central nervous system depressant effects. **Direction:** The combination has primary risks that include increased sedation and drowsiness; both drugs can prolong the QT interval, potentially increasing serious cardiac arrhythmia risk; enhanced dopamine blockade increases extrapyramidal symptom risk. **Consequence:** Increased sedation and drowsiness may impair cognitive function and increase fall risk, especially in elderly patients; potential increased risk of serious cardiac arrhythmias; elevated risk of extrapyramidal symptoms such as tardive dyskinesia, dystonia, and parkinsonism. **Evidence:** General web search (Tavily) [inferred] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene EPM2A. Haloperidol EPM2A (CC): efficacy (PMID:27092952); Quetiapine EPM2A (CC): efficacy (PMID:27092952) **Hyperlink:** General web search (Tavily)
	Haloperidol + Sertraline **Findings:** Sertraline is listed as a CYP2D6 inhibitor that can increase haloperidol plasma concentrations when coadministered with haloperidol. The label warns that increased haloperidol concentrations may increase adverse events, including QTc interval prolongation, and recommends monitoring for increased or prolonged haloperidol effects and decreasing the haloperidol injection dose as needed. **Mechanism:** Sertraline is listed as a CYP2D6 inhibitor; haloperidol metabolism involves CYP2D6 and CYP3A4. **Direction:** Increased haloperidol plasma concentrations. **Consequence:** Increased risk of adverse events, including QTc interval prolongation. **Evidence:** DailyMed label (Drug Interactions section) [curated] **Hyperlink:** DailyMed
	Haloperidol + Venlafaxine **Findings:** Venlafaxine is listed as a CYP2D6 inhibitor whose coadministration with haloperidol increased haloperidol plasma concentrations. The label states that increased haloperidol concentrations may increase adverse events, including QTc interval prolongation, and recommends monitoring for increased or prolonged haloperidol effects and decreasing the haloperidol injection dose as needed. **Mechanism:** Venlafaxine is listed as a CYP2D6 inhibitor; CYP2D6 is involved in haloperidol metabolism. **Direction:** Coadministration increased haloperidol plasma concentrations. **Consequence:** Increased haloperidol plasma concentrations may increase adverse events, including QTc interval prolongation. **Evidence:** DailyMed label (Drug Interactions section) [curated]; DailyMed label (Drug Interactions section) [curated] **Hyperlink:** DailyMed

LAMOTRIGINE

POSITIVE IMPACT	NEGATIVE IMPACT
Lamotrigine + Levetiracetam **Findings:** Levetiracetam did not influence lamotrigine plasma concentrations, and lamotrigine did not influence levetiracetam pharmacokinetics in placebo-controlled clinical studies of antiepileptic drugs. **Direction:** no effect on lamotrigine plasma concentration or levetiracetam pharmacokinetics **Consequence:** no clinically meaningful pharmacokinetic interaction was indicated **Evidence:** DailyMed label (Drug Interactions section) [curated]; General web search (Tavily) [inferred] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene SCN1A. Lamotrigine SCN1A (CT): efficacy (PMID:28753467); Levetiracetam SCN1A (CT): efficacy (PMID:28753467)	**Lamotrigine + Oxcarbazepine** **Findings:** Lamotrigine and oxcarbazepine coadministration did not significantly affect lamotrigine AUC or Cmax, nor did lamotrigine significantly affect MHD AUC or Cmax. However, headache, dizziness, nausea, and somnolence occurred more frequently with the lamotrigine–oxcarbazepine combination than with monotherapy. **Direction:** No significant pharmacokinetic change; adverse events occurred more frequently with combination therapy. **Consequence:** Headache, dizziness, nausea, and somnolence occurred more frequently during combination therapy than during monotherapy. **Evidence:** General web search (Tavily) [inferred] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene SCN1A. Lamotrigine SCN1A (CT): efficacy

POSITIVE IMPACT	NEGATIVE IMPACT
Hyperlink: DailyMed	(PMID:28753467); Oxcarbazepine SCN1A (CT): efficacy (PMID:28753467) **Hyperlink:** General web search (Tavily)
Lamotrigine + Olanzapine **Findings:** Lamotrigine and olanzapine can safely be combined in healthy volunteers at the low doses studied, without a clinically significant interaction; olanzapine did not significantly affect lamotrigine AUC0–24h or plasma concentrations. Lamotrigine Tmax was significantly prolonged during olanzapine coadministration, possibly because of olanzapine's anticholinergic properties, and mild sedation was the only adverse effect reported. **Mechanism:** possibly because of the anticholinergic properties associated with olanzapine **Direction:** olanzapine did not significantly affect lamotrigine disposition; lamotrigine Tmax was significantly prolonged during olanzapine coadministration **Consequence:** mild sedation was the only adverse effect that occurred during lamotrigine and olanzapine coadministration **Evidence:** General web search (Tavily) [inferred] **Hyperlink:** General web search (Tavily)	**Lamotrigine + Phenytoin** **Findings:** Phenytoin decreases lamotrigine concentrations by approximately 40% according to the lamotrigine DailyMed Drug Interactions section. **Mechanism:** Phenytoin is listed among drugs that decrease lamotrigine concentrations; the label identifies UGT enzymes as responsible for lamotrigine metabolism. **Direction:** Decreased lamotrigine concentration approximately 40%. **Evidence:** DailyMed label (Drug Interactions section) [curated]; General web search (Tavily) [inferred] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene CYP3A4. Lamotrigine CYP3A4 (TT): reduced efficacy (PMID:28343093); Phenytoin CYP3A4 (TT): reduced efficacy (PMID:28343093) **Hyperlink:** DailyMed
Lamotrigine + Sertraline **Findings:** Lamotrigine dose-corrected concentrations were slightly higher with sertraline than without sertraline (60.4 vs 51.1 µmol/L × 1,000/mg/day; p=0.42), and the authors concluded the slower lamotrigine metabolism was not believed to be clinically significant. They cautioned that limited power may have overlooked a clinically relevant difference. **Mechanism:** slightly slower metabolism of lamotrigine in patients receiving lamotrigine with sertraline **Direction:** dose-corrected lamotrigine concentration was 60.4 vs 51.1 µmol/L × 1,000/mg/day with vs without sertraline (p=0.42) **Consequence:** not believed to be of clinical significance; limited power may have overlooked a difference that could be clinically relevant **Evidence:** External evidence (PubMed) [inferred] **Hyperlink:** PubMed	**Lamotrigine + Valproic Acid** **Findings:** Valproic acid co-administration can affect lamotrigine pharmacokinetics by inhibiting lamotrigine metabolism or by protein-binding displacement. The label indicates increased monitoring of valproate and concomitant drug concentrations and dosage adjustment when interacting drugs are introduced or withdrawn. **Mechanism:** Valproic acid can affect lamotrigine pharmacokinetics by inhibiting metabolism or protein-binding displacement. **Direction:** can affect lamotrigine pharmacokinetics **Evidence:** DailyMed label (Drug Interactions section) [curated] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene CYP3A4. Lamotrigine CYP3A4 (TT): reduced efficacy (PMID:28343093); Valproic Acid CYP3A4 (TT): reduced efficacy (PMID:28343093) **Hyperlink:** DailyMed

LEVETIRACETAM

POSITIVE IMPACT	NEGATIVE IMPACT
Levetiracetam + Phenytoin **Findings:** Levetiracetam (3000 mg daily) had no effect on the pharmacokinetic disposition of phenytoin in patients with refractory epilepsy, and phenytoin did not affect levetiracetam pharmacokinetics. **Direction:** no effect on pharmacokinetic disposition of phenytoin; levetiracetam pharmacokinetics not affected by phenytoin **Evidence:** DailyMed label (Drug Interactions section) [curated]; General web search (Tavily) [inferred] **Hyperlink:** DailyMed	**Levetiracetam + Olanzapine** **Findings:** Levetiracetam combined with olanzapine was associated with a significantly greater reduction in YMRS scores than placebo plus olanzapine at week 16 (mean 2.03 vs 16.63, P < 0.001) in acute mania, while lethargy occurred more frequently with levetiracetam (23.3% vs 3.3%, P = 0.05). The authors concluded the combination may benefit bipolar disorder patients unresponsive to olanzapine alone, with further research needed to confirm the findings. **Direction:** greater reduction in YMRS scores; more frequent lethargy **Evidence:** External evidence (PubMed) [inferred] **Hyperlink:** PubMed
Levetiracetam + Quetiapine **Findings:** Adjunctive levetiracetam plus quetiapine significantly improved acute mania outcomes versus placebo plus quetiapine: YMRS scores showed a significant time × treatment interaction (P = 0.04), greater reduction by week 4 (P = 0.045), and better treatment response (P = 0.046). The retrieved abstract states the trial evaluated safety and tolerability but is truncated before any pair-specific safety/tolerability results are reported. **Direction:** Improved acute mania outcomes with levetiracetam plus quetiapine versus placebo plus quetiapine. **Consequence:** Greater reduction in YMRS scores and better response to treatment. **Evidence:** External evidence (PubMed) [inferred] **Hyperlink:** PubMed	
Levetiracetam + Valproic Acid **Findings:** Levetiracetam did not alter the pharmacokinetics of valproate in healthy volunteers, and valproate did not modify the rate or extent of levetiracetam absorption or exposure in the label's clinical pharmacokinetic assessment. **Mechanism:** Levetiracetam is unlikely to produce or be subject to pharmacokinetic interactions and does not affect the in vitro glucuronidation of valproic acid. **Direction:** Levetiracetam did not alter valproate pharmacokinetics; valproate did not modify levetiracetam rate or extent of absorption or exposure. **Consequence:** No pharmacokinetic interaction was found for levetiracetam and valproate in the label's assessment. **Evidence:** DailyMed label (Drug Interactions section) [curated] **Hyperlink:** DailyMed	

LITHIUM

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	**Lithium + Paroxetine** **Findings:** Paroxetine's label states that concomitant use of paroxetine with serotonergic drugs, including lithium, increases the risk of serotonin syndrome. It recommends monitoring for signs and symptoms of serotonin syndrome, particularly during treatment initiation and dosage increases, and considering discontinuation of paroxetine and/or the concomitant serotonergic drug if serotonin syndrome occurs. **Mechanism:** concomitant use of serotonergic drugs with paroxetine **Direction:** increases the risk **Consequence:** serotonin syndrome **Evidence:** DailyMed label (Drug Interactions section) [curated] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene HTR1B. Lithium HTR1B (TA): efficacy (PMID:21449676); Paroxetine HTR1B (TA): toxicity (PMID:21449676) **Hyperlink:** DailyMed **Lithium + Sertraline** **Findings:** Sertraline's label states that concomitant use of serotonergic drugs, including lithium, with sertraline hydrochloride increases the risk of serotonin syndrome. It recommends monitoring for signs and symptoms of serotonin syndrome, particularly during treatment initiation and dosage increases, and considering discontinuation of sertraline and/or the concomitant serotonergic drug if serotonin syndrome occurs. **Mechanism:** concomitant serotonergic drugs with sertraline hydrochloride **Direction:** increases the risk **Consequence:** serotonin syndrome **Evidence:** DailyMed label (Drug Interactions section) [curated] **Hyperlink:** DailyMed **Lithium + Venlafaxine** **Findings:** Lithium did not affect steady-state pharmacokinetics of venlafaxine or O-desmethylvenlafaxine, and venlafaxine had no effect on lithium pharmacokinetics in the DailyMed label study. The label also lists lithium among serotonergic drugs with venlafaxine, warning of a potential for serotonin syndrome and advising careful patient observation. **Mechanism:** serotonergic drugs **Direction:** potential for serotonin syndrome; no pharmacokinetic effect reported **Consequence:** potential for serotonin syndrome **Evidence:** DailyMed label (Drug Interactions section) [curated]; General web search (Tavily) [inferred] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene FKBP5, HTR1B. Lithium FKBP5 (TC): toxicity (PMID:21449676); Venlafaxine FKBP5 (TC): toxicity (PMID:21449676); Lithium HTR1B (TA): efficacy (PMID:21449676); Venlafaxine HTR1B (TA): efficacy (PMID:21449676) **Hyperlink:** DailyMed

METHYLPHENIDATE

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	**Methylphenidate + Phenytoin** **Findings:** Methylphenidate is listed in phenytoin's label among drugs that may increase phenytoin serum levels. The label indicates this interaction may require attention to achieve optimal clinical outcome. **Direction:** may increase phenytoin serum levels **Evidence:** DailyMed label (Drug Interactions section) [curated]; General web search (Tavily) [inferred] **Hyperlink:** DailyMed **Methylphenidate + Quetiapine** **Findings:** Adding quetiapine to OROS methylphenidate improved ADHD symptoms, aggression, and global functioning in adolescents insufficiently responsive to methylphenidate alone; 42% met all improvement criteria and 79% had minimal aggression. Mild transient sedation occurred in about half, and weight gain during combination treatment offset prior methylphenidate-related weight loss. **Direction:** improved ADHD symptoms, aggression, and global functioning; mild transient sedation; weight gain offset prior weight loss **Consequence:** 42% met all criteria for clinically significant improvement and 79% showed minimal aggression; mild and transient sedation was reported by about half the cases **Evidence:** External evidence (PubMed) [inferred] **Hyperlink:** PubMed

NORTRIPTYLINE

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	**Nortriptyline + Paroxetine** **Findings:** Paroxetine, an SSRI, inhibits CYP2D6 and the nortriptyline label warns that concomitant use of TCAs with CYP2D6-inhibiting drugs may cause abrupt TCA toxicity, with SSRI–TCA interaction size potentially ranging up to an 8-fold increase in TCA plasma AUC. The label indicates caution with coadministration of TCAs and SSRIs and states that lower TCA doses may be required. **Mechanism:** Paroxetine inhibits cytochrome P450 2D6; nortriptyline is a tricyclic antidepressant affected by CYP2D6 inhibition. **Direction:** TCA plasma AUC may increase, potentially up to 8-fold. **Consequence:** A patient stable on a TCA may become abruptly toxic when given a CYP2D6-inhibiting drug concomitantly. **Evidence:** DailyMed label (Drug Interactions section) [curated] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene BDNF. Nortriptyline BDNF (TC): toxicity (PMID:19641488); Paroxetine BDNF (CT): reduced efficacy (PMID:24577123) **Hyperlink:** DailyMed **Nortriptyline + Sertraline** **Findings:** Nortriptyline's label states that sertraline, as an SSRI, inhibits CYP2D6 and that coadministration of TCAs such as nortriptyline with SSRIs warrants caution because CYP2D6 inhibition may cause a stable TCA patient to become abruptly toxic; TCA plasma concentration increases may be small or as large as an 8-fold AUC increase. **Mechanism:** Sertraline, as an SSRI, inhibits cytochrome P450 2D6, which is relevant to tricyclic antidepressants such as nortriptyline. **Direction:** TCA plasma concentrations may increase, with increases described as small or as large as an 8-fold increase in TCA AUC. **Consequence:** A patient stable on a TCA may become abruptly toxic when given a CYP2D6-inhibiting drug as concomitant therapy. **Evidence:** DailyMed label (Drug Interactions section) [curated] **Hyperlink:** DailyMed **Nortriptyline + Valproic Acid** **Findings:** Valproic Acid's label states that dosage adjustment of amitriptyline/nortriptyline may be necessary if used concomitantly with Valproic Acid Capsules. **Evidence:** DailyMed label (Drug Interactions section) [curated] **Hyperlink:** DailyMed

OLANZAPINE

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	No pairs currently fall in this category.

OXCARBAZEPINE

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	**Oxcarbazepine + Perampanel** **Findings:** Concomitant oxcarbazepine with perampanel decreased perampanel plasma levels by approximately 50% to 67% because oxcarbazepine is a moderate or strong CYP3A4 inducer. The label recommends increasing perampanel starting doses in the presence of CYP3A4 inducers and closely monitoring clinical response and tolerability when these inducers are introduced or withdrawn; perampanel dose adjustment may be necessary. **Mechanism:** Oxcarbazepine is included among moderate and strong CYP3A4 inducers that increase clearance of perampanel. **Direction:** Decreased perampanel plasma levels by approximately 50% to 67%. **Consequence:** Reduced perampanel plasma concentrations, with need to monitor clinical response and tolerability. **Evidence:** DailyMed label (Drug Interactions section) [curated]; General web search (Tavily) [inferred] **Hyperlink:** DailyMed **Oxcarbazepine + Phenytoin** **Findings:** Coadministration of oxcarbazepine (Oxtellar XR) with phenytoin decreased blood levels of oxcarbazepine's active metabolite (MHD), and the label recommends monitoring MHD during titration because Oxtellar XR dosage adjustment may be required. The label also recommends monitoring phenytoin plasma levels during Oxtellar XR titration or dose changes because a decrease in phenytoin dosage may be required.

POSITIVE IMPACT	NEGATIVE IMPACT
	Mechanism: Phenytoin is listed as a strong CYP3A4 inducer or UGT inducer affecting oxcarbazepine's active metabolite MHD. **Direction:** Coadministration decreased blood levels of oxcarbazepine's active metabolite MHD; phenytoin dosage decrease may be required during Oxtellar XR titration or dosage modification. **Consequence:** Altered plasma levels requiring monitoring and possible dosage adjustment. **Evidence:** DailyMed label (Drug Interactions section) [curated]; DailyMed label (Drug Interactions section) [curated]; General web search (Tavily) [inferred] **Hyperlink:** DailyMed

PAROXETINE

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	**Paroxetine + Sertraline** **Findings:** Paroxetine and sertraline have additive serotonergic activity when combined, increasing the risk of serotonin syndrome, activation, sleep disturbance, and agitation. **Mechanism:** Additive serotonergic activity from combined serotonergic agents. **Direction:** Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. **Consequence:** Increased serotonin-syndrome risk, activation, sleep disturbance, agitation. **Evidence:** Pharmacodynamic interaction reference [curated]; General web search (Tavily) [inferred] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene SRP19. Paroxetine SRP19 (GG): efficacy (PMID:22795047); Sertraline SRP19 (GG): efficacy (PMID:22795047) **Paroxetine + Venlafaxine** **Findings:** Paroxetine may increase venlafaxine exposure because paroxetine is a CYP2D6 inhibitor and venlafaxine is listed as a CYP2D6 substrate. The label recommends decreasing the dosage of the CYP2D6 substrate if needed during concomitant paroxetine use, and increasing it may be needed if paroxetine is discontinued. **Mechanism:** Paroxetine is a CYP2D6 inhibitor; venlafaxine is listed as a CYP2D6 substrate. **Direction:** Concomitant use of paroxetine with a CYP2D6 substrate may increase exposure of the CYP2D6 substrate. **Consequence:** May increase venlafaxine exposure. **Evidence:** DailyMed label (Drug Interactions section) [curated] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene HTR1B, HTR2A, HTR7. Paroxetine HTR1B (TA): toxicity (PMID:21449676); Venlafaxine HTR1B (TA): efficacy (PMID:21449676); Paroxetine HTR2A (TT): toxicity (PMID:16710319); Venlafaxine HTR2A (AG): efficacy (PMID:22006095); Paroxetine HTR7 (GG): efficacy (PMID:30874608); Venlafaxine HTR7 (GG): efficacy (PMID:30874608) **Hyperlink:** DailyMed

PERAMPANEL

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	**Perampanel + Phenytoin** **Findings:** Concomitant phenytoin with perampanel decreased perampanel plasma levels by approximately 50 to 67% because phenytoin is a moderate or strong CYP3A4 inducer. The label recommends increased starting doses for perampanel in the presence of such inducers and close monitoring for clinical response and tolerability when they are introduced or withdrawn; perampanel dose adjustment may be necessary. **Mechanism:** Phenytoin is included among moderate and strong CYP3A4 inducers that increase clearance of perampanel. **Direction:** Decreased perampanel plasma levels by approximately 50 to 67%. **Consequence:** Reduced perampanel exposure; patients should be monitored for clinical response and tolerability. **Evidence:** DailyMed label (Drug Interactions section) [curated] **Hyperlink:** DailyMed

PHENYTOIN

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	**Phenytoin + Quetiapine** **Findings:** Phenytoin, a CYP3A4 inducer, increased the mean oral clearance of quetiapine 5-fold; the label states quetiapine exposure is decreased by CYP3A4

POSITIVE IMPACT	NEGATIVE IMPACT
	inducers. Increased quetiapine doses up to 5-fold may be required to maintain control of schizophrenia symptoms during concomitant phenytoin, and quetiapine should be reduced to the original dose within 7–14 days after the inducer is discontinued. **Mechanism:** Phenytoin is a CYP3A4 inducer that increases quetiapine oral clearance and decreases quetiapine exposure. **Direction:** Mean oral clearance of quetiapine increased 5-fold; quetiapine exposure is decreased. **Consequence:** Reduced quetiapine exposure may require higher quetiapine doses to maintain control of symptoms of schizophrenia. **Evidence:** DailyMed label (Drug Interactions section) [curated]; General web search (Tavily) [inferred] **Hyperlink:** DailyMed **Phenytoin + Sertraline** **Findings:** Sertraline is listed among antidepressants that may increase phenytoin serum levels. The phenytoin label states that adding or withdrawing such agents in patients on phenytoin therapy may require phenytoin dose adjustment to achieve optimal clinical outcome. **Direction:** Sertraline may increase phenytoin serum levels. **Evidence:** DailyMed label (Drug Interactions section) [curated]; DailyMed label (Drug Interactions section) [curated]; External evidence (PubMed) [inferred] **Hyperlink:** DailyMed **Phenytoin + Valproic Acid** **Findings:** Phenytoin's label states that valproate/valproic acid may either increase or decrease phenytoin serum levels. Because the label identifies bidirectional phenytoin-level changes with concomitant valproic acid, this interaction may require clinical attention to phenytoin exposure. **Direction:** Valproate/valproic acid may either increase or decrease phenytoin serum levels. **Consequence:** Altered phenytoin serum levels **Evidence:** DailyMed label (Drug Interactions section) [curated]; DailyMed label (Drug Interactions section) [curated] **Patient PGx context [predicted] (not part of this verdict):** both drugs carry a finding for the shared gene CYP1A1, CYP3A4, GABRA1. Phenytoin CYP1A1 (CA): toxicity (PMID:21121773); Valproic Acid CYP1A1 (CA): toxicity (PMID:21121773); Phenytoin CYP3A4 (TT): reduced efficacy (PMID:28343093); Valproic Acid CYP3A4 (TT): reduced efficacy (PMID:28343093); Phenytoin GABRA1 (GA): reduced efficacy (PMCID:PMC3125053); Valproic Acid GABRA1 (GA): reduced efficacy (PMCID:PMC3125053) **Hyperlink:** DailyMed

QUETIAPINE

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	**Quetiapine + Valproic Acid** **Findings:** General web evidence reports that valproic acid could increase quetiapine exposure; one clinical study found a 77% rise in quetiapine plasma concentrations, while another found 17% higher Cmax but no AUC change. Clinical significance was unknown, but case reports described cervical dystonia with high quetiapine after valproic acid was added and Parkinsonism during concomitant treatment. **Direction:** Valproic acid could increase quetiapine exposure; 77% rise in plasma concentrations in one study, 17% higher Cmax but no AUC change in another. **Consequence:** Clinical significance unknown; case reports described cervical dystonia with high quetiapine after valproic acid was added and Parkinsonism during concomitant treatment. **Evidence:** General web search (Tavily) [inferred] **Hyperlink:** General web search (Tavily)

SERTRALINE

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	**Sertraline + Venlafaxine** **Findings:** Sertraline hydrochloride may increase the exposure of CYP2D6 substrates, including venlafaxine, because sertraline hydrochloride is a CYP2D6 inhibitor. The label recommends decreasing the dosage of the CYP2D6 substrate if needed during concomitant sertraline use and increasing it if needed when sertraline is discontinued. **Mechanism:** Sertraline hydrochloride is a CYP2D6 inhibitor; venlafaxine is listed as a CYP2D6 substrate example. **Direction:** May increase exposure of the CYP2D6 substrate. **Consequence:** Increased exposure of venlafaxine as a CYP2D6 substrate may occur with concomitant sertraline hydrochloride use.

POSITIVE IMPACT	NEGATIVE IMPACT
	Evidence: DailyMed label (Drug Interactions section) [curated] **Hyperlink:** DailyMed

VALPROIC ACID

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	No pairs currently fall in this category.

VENLAFAXINE

POSITIVE IMPACT	NEGATIVE IMPACT
No pairs currently fall in this category.	No pairs currently fall in this category.

Known Limitations (Drug Interaction Flag Sheet):

- A pair with no finding from Flockhart, pharmacodynamic rules, PubMed, DailyMed, ClinicalTrials.gov, or Tavily is dropped from this report entirely, not shown as a negative/no-concern finding -- 'not listed here' means 'no evidence was found for this exact pair', never 'confirmed safe'.
- A DailyMed 'not mentioned in reviewed label section' result is never itself a finding -- it means this module checked the drug's real DRUG INTERACTIONS label section and the other drug's name was not there, not that no interaction exists; it only ever appears as supplementary context alongside whatever other evidence a pair has.
- Tavily (general web search) is queried directly and unconditionally alongside PubMed, DailyMed, and ClinicalTrials.gov for every pair here -- unlike elsewhere in this codebase (external_evidence_summary.py's question-answering paths), it is not gated behind the other sources returning no_results first. Every Tavily-sourced evidence entry is labeled 'General web search (Tavily)' so it is never mistaken for a dedicated medical/regulatory source, and it is always the lowest-priority candidate for a pair's primary Finding text.
- A source whose language does not clearly match an explicit positive-outcome or negative-outcome keyword never resolves a pair on its own -- this module falls through to the next candidate source in priority order instead, and if no candidate ever resolves clearly, the pair is dropped from the report entirely, the same as a pair with no evidence at all. There is no 'Needs Review' column here; most real titles/excerpts are topic descriptions, not verdicts, and this module never guesses which way an unclear one points.
- CPIC and ClinVar (gene-level sources) are not re-queried per pair here -- see the module docstring for why folding a single drug's own genotype-predicted finding into a pair's verdict here would conflate two different questions. Where this patient's genetics panel covers a screened drug, its real CPIC/ClinVar findings are in the Pharmacogenomics section, not duplicated here.
- Flockhart Cytochrome P450 Drug Interaction Table (curated-snapshot-2024) curated CYP-pathway data covers only a small hand-curated set of drugs; a pair outside that set relies on the other five sources, not a gap in this report.
- Positive/Negative Impact and the Finding come from the Finding Agent reading each pair's own source evidence (see classification_method); a title, snippet or keyword hit alone never decides a pair. Animal/cell studies, comparator-only studies, retracted papers and studies involving a third drug are excluded (source_rejection_counts shows how many of each). Patient PGx findings are shown as context only and never classify a pair.

Current Regimen — Curated-Source Coverage

MEDICINE	STATUS	MEANING	ACTION
Lamotrigine	COVERED	Curated CYP/pharmacokinetic reference data exists for this drug (Flockhart Cytochrome P450 Drug Interaction Table). This is reference-data coverage, not a pair's resolution.	No additional action required beyond the current-current pair evidence below.
Levetiracetam (Keppra)	COVERED	Curated CYP/pharmacokinetic reference data exists for this drug (Flockhart Cytochrome P450 Drug Interaction Table). This is reference-data coverage, not a pair's resolution.	No additional action required beyond the current-current pair evidence below.

Clinical inference: Absence of a resolved current-current interaction does not establish absence of interaction. Unresolved predicted-model coverage limits interpretation; the active medication list should be reassessed against a complete interaction source before any treatment change. The therapy-level cards below give the supporting, counter, and unresolved evidence behind that conclusion for each current medication, across pharmacokinetic, pharmacodynamic, pharmacogenomic, clinical, EEG, and laboratory sources -- not a pass/fail interaction check.

- SuperCYPsPred predicted pharmacokinetic evidence is not available locally for any drug in this regimen.
- No therapeutic drug concentration data is present in the source reports.
- No renal function (eGFR) data is present in the source reports.

Candidate / Proposed Medication-Pair Screening

No candidate or proposed medications are present in this patient's regimen; no candidate-pair screening applies.

Lamotrigine EFFECTIVENESS INCOMPLETE

Therapeutic evidence present, but clinical effectiveness remains incomplete.

Supporting Evidence

- CBC status across the observation window: stable within reference ranges; no documented hematological toxicity. — digital_twin_state
- No documented drug toxicity for this agent in the clinical-notes source. — clinical_inference.medication_response
- SCN1A (CT): efficacy — findings_by_therapeutic_class

Counter-Evidence

- Breakthrough seizures persisted on the current regimen, with a recorded peak of 11 episodes on 2026-02-15. — clinical_inference.seizure_analysis
- ABCB1 (GG): reduced efficacy — findings_by_therapeutic_class
- CYP3A4 (TT): reduced efficacy — findings_by_therapeutic_class

Clinical Impression

4 supporting, 5 counter, and 13 unresolved evidence item(s) were identified for Lamotrigine. Clinical-notes inference: Current antiseizure therapy is associated with stable clinician impressions but incomplete seizure suppression.

Recommended Monitoring

- Continue EEG-derived seizure-risk monitoring.
- Continue periodic CBC and safety-laboratory monitoring.
- Continue seizure frequency, duration, and clustering tracking.
- Review the priority safety flags in the Pharmacogenomics section before adding or changing an interacting medication.

Levetiracetam (Keppra) EFFECTIVENESS INCOMPLETE

Therapeutic evidence present, but clinical effectiveness remains incomplete.

Supporting Evidence

- CBC status across the observation window: stable within reference ranges; no documented hematological toxicity. — digital_twin_state
- No documented drug toxicity for this agent in the clinical-notes source. — clinical_inference.medication_response
- SCN1A (CT): efficacy — findings_by_therapeutic_class

Counter-Evidence

- Breakthrough seizures persisted on the current regimen, with a recorded peak of 11 episodes on 2026-02-15. — clinical_inference.seizure_analysis
- Seizure clustering was documented during the observation window. — clinical_inference.seizure_analysis
- EEG synthesis remained consistent with active epilepsy (EEG-derived risk: Elevated; future seizure probability: Elevated). — digital_twin_state

Clinical Impression

3 supporting, 3 counter, and 7 unresolved evidence item(s) were identified for Levetiracetam (Keppra). Clinical-notes inference: Current antiseizure therapy is associated with stable clinician impressions but incomplete seizure suppression.

Recommended Monitoring

- Continue EEG-derived seizure-risk monitoring.
- Continue periodic CBC and safety-laboratory monitoring.
- Continue seizure frequency, duration, and clustering tracking.
- Review the priority safety flags in the Pharmacogenomics section before adding or changing an interacting medication.

Limitations of this screen: This assessment is clinical decision support, not an automated prescribing system; no dose, start, or stop decision is made or implied here. No proposed/candidate medication list is present in the source data, so only the current regimen's pairwise and therapy-level evidence is evaluated. Curated pharmacokinetic (CYP) coverage is limited to a small bundled reference table; a drug not listed there is reported as unresolved, not as free of interactions. Therapy assessments also draw on the live external-evidence layer (see Known Limitations under Pharmacogenomics, above, for its source-by-source scope and caveats).

07 Pharmacogenomics

76 variants across 47 medications · reported 10 Sep 2026

Pharmacogenomic Panel Summary

CYP2D6 **GG**	CYP2C19 **GA**	CYP2C9 ***1/*3 - Intermediate Metabolizer**	CYP2B6 ***1/*6 - Intermediate Metabolizer**	CYP3A5 ***3/*3 - Poor Metabolizer**
CYP1A2 **AA**	CYP3A4 **TT**	UGT1A4 ***1/*3A - Decreased Function**	POR ***1/*28 - Decreased Function**	

Interpretive method: a pharmacogenomic association is weighed against this patient's own observed clinical response, laboratory, and EEG evidence -- it is never used as an isolated prescribing instruction, and a favorable genotype cannot override documented toxicity or persistent symptoms. The table below shows the single most clinically significant finding per current-regimen drug; the full therapy-level assessment -- this finding combined with clinical, EEG, laboratory, and external-database evidence -- is presented in Section 06, Drug Interactions.

Patient-Specific PGx Findings

THERAPY	GENE/GENOTYPE	PATIENT-SPECIFIC INTERPRETATION	CLINICAL RELEVANCE
Lamotrigine	CYP3A4 (TT)	Genotype is associated with a reduced-efficacy response to this therapy.	Moderate relevance; flagged as clinically significant in the source panel (citation: PMID:28343093).
Levetiracetam	SCN1A (CT)	Genotype is associated with a favorable efficacy response to this therapy.	Supportive relevance; flagged as clinically significant in the source panel (citation: PMID:28753467).

External Evidence Verification

Database evidence only, not patient-verified: each block below is a live public-database lookup for the same gene/drug pair as the patient-specific finding in Table 1 above -- it establishes general scientific plausibility for that pair, not a fact confirmed for this individual patient. An unresolved result means this exact query returned nothing, or failed, at the time checked -- it is not evidence that no relevant data exists, and must not be read as a clean or reassuring negative.

- **CYP3A4 → Lamotrigine** PARTIALLY VERIFIED
  - CPIC: Unresolved. ClinVar: Uncertain significance.
  - PubMed: PMID 21635243 found. Overall: Partially resolved (2 of 3 sources).
- **SCN1A → Levetiracetam** PARTIALLY VERIFIED
  - CPIC: Unresolved. ClinVar: Uncertain significance.
  - PubMed: PMID 39200163 found. Overall: Partially resolved (2 of 3 sources).

Provenance

- **CYP3A4 → Lamotrigine:** ClinVar — Accession VCV004885026 (70 total gene-wide records) · LIVE · Retrieved 11 Sep 2026; PubMed — PMID 21635243 · LIVE · Retrieved 11 Sep 2026. Note: Table 1's citation is from the patient's own PGx report; this is an independent live PubMed search, may differ.
- **SCN1A → Levetiracetam:** ClinVar — Accession VCV004887952 (5406 total gene-wide records) · LIVE · Retrieved 11 Sep 2026; PubMed — PMID 39200163 · LIVE · Retrieved 11 Sep 2026. Note: Table 1's citation is from the patient's own PGx report; this is an independent live PubMed search, may differ.

Evidence Scope: live-verified via ClinVar, CPIC, DailyMed, PubMed, and ClinicalTrials.gov only. The rest of the NeuroTwin resource catalog (PharmGKB, DrugBank, Reactome, STRING, GTEx, and others) is out of scope for this live-lookup layer and is not reflected anywhere in this table — see Known Limitations below.

Known Limitations (External Evidence):

- This module queries exactly five public APIs (PubMed, ClinVar, DailyMed, CPIC, ClinicalTrials.gov); it does not cover the rest of the NeuroTwin resource catalog (Reactome, STRING, GTEx, RCSB PDB, etc.) -- those remain out of scope for this module and must not be inferred from it.
- A 'no_results' status means the API returned zero matching records for the exact query issued; it is not evidence of absence, and must not be reported as a negative finding.
- Responses may be served from the on-disk cache in reports/external_evidence/.cache/ rather than a fresh call; each record's 'from_cache' and 'retrieved_at' fields state which. Pass --refresh to force live calls.

08 EEG

Longitudinal interictal / preictal / ictal signal characteristics

35	12	11	12
Total Recordings	Interictal	Preictal	Ictal

Signal Variance Midpoint by Phase (μV²)



EEG-derived risk: Elevated

Future seizure probability: Elevated

Dominant feature: Signal Variance

Phase Analysis

PHASE	STATE	ACTIVITY	VARIABILITY
Interictal	Baseline	Stable	Low
Preictal	Transition	Increasing Instability	Moderate-High
Ictal	Seizure	Hypersynchronous	Very High

PHASE	AMPLITUDE RANGE	STD DEV RANGE	VARIANCE RANGE
Interictal	-89; 99	11.68; 23.5	136.44; 552.46
Preictal	-345; 440	14.48; 92.07	209.71; 8477.48
Ictal	-504; 784	52.5; 286.42	2756.47; 82033.96

Longitudinal EEG recordings demonstrate a consistent transition from stable interictal activity to preictal electrical instability followed by high-amplitude ictal seizure activity. Findings are consistent with the dataset's interictal/preictal/ictal phase taxonomy, indicating reproducible seizure-related signal evolution; this is a signal-classification observation, not a clinician-verified diagnosis.

09 MRI / CT

Structural imaging: neuroimaging key findings

CT — 6 studies, 12 Jan – 29 Jun 2026

DATE	IMPRESSION
12 Jan 2026	No acute intracranial abnormality is evident on this single-image model-assisted review.
16 Feb 2026	No gross acute abnormality is evident on this limited skull-base level image; assessment is limited by slice level and image quality.
11 Mar 2026	No acute intracranial abnormality is evident on this single-image model-assisted review.
20 Apr 2026	No acute intracranial abnormality is evident on this high-convexity image; ventricular assessment is not available at this slice level.
14 May 2026	No acute intracranial abnormality is evident on this limited-quality near-vertex image.
29 Jun 2026	No acute intracranial abnormality is evident. Incidental bilateral choroid plexus calcifications are present.

Across the available CT image files, no gross acute intracranial hemorrhage, midline shift, or mass effect is evident on this model-assisted image-only review. Direct progression assessment is limited because the files represent single slices with different slice levels, resolutions, and image quality. This output is not a radiologist-verified diagnostic report.

MRI — Metadata-supported structural stability, diagnostic interpretation not available

FINDING	STATUS	DETAIL
Brain Morphology	NOT DIRECTLY ASSESSED	The provided files include volumetric MRI metadata, but no radiologist interpretation or morphometric segmentation was available for direct morphology assessment.
Brain Volume	STABLE WITHIN AVAILABLE METADATA	T1w nonzero volume range: 8387642 to 8777268 mm3 (mean 8641065 mm3; range 4.5%). FLAIR nonzero volume range: 6022385 to 6625690 mm3 (mean 6231344 mm3; range 9.7%). These are preprocessing-derived nonzero voxel volumes, not formal brain volumetry.
Tissue Signal Characteristics	QUANTITATIVELY SUMMARIZED, NOT DIAGNOSTICALLY INTERPRETED	FLAIR mean intensity 28.76 to 32.27; standard deviation 47.57 to 54.16; T1w mean intensity 99.82 to 105.56; standard deviation 153.13 to 162.05. Intensity statistics are useful for preprocessing consistency but are not diagnostic tissue-signal interpretation.

Not directly assessed from the available imaging metadata (this means "not evaluated," not "abnormal"): White Matter, Hippocampus, Cortical Structure, Mass Lesion, Hemorrhage, Infarction.

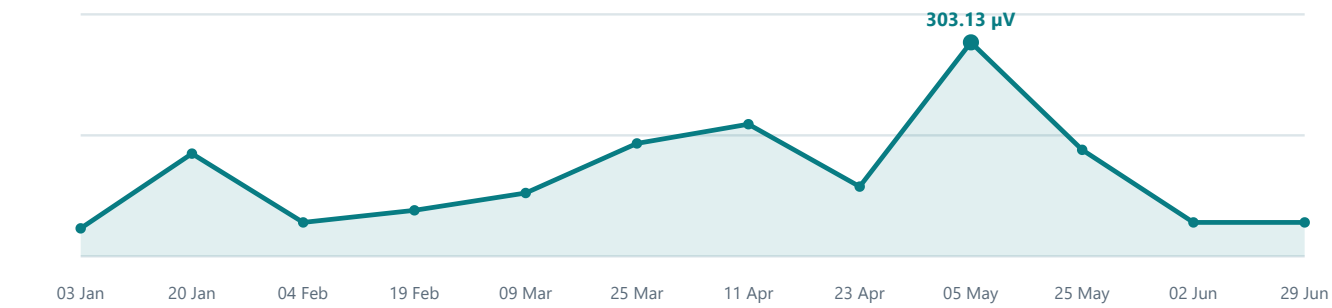
Stable by available preprocessing metadata. 6 MRI series were available across 3 study identifiers, including 3D T1-weighted MRI, T2-FLAIR MRI. Preprocessed metadata show consistent 3D acquisition characteristics and no metadata-level evidence of major global structural progression. A radiology report, lesion segmentation, hippocampal volumetry, and cortical thickness analysis were not available, so subtle pathology cannot be excluded from these metadata alone.

10 ECG

single channel ("ECG SD") in every recording

12	99h 11m	256
Recordings	Total Duration	Sampling Rate (Hz)

Amplitude Std Dev by Recording Date (µV)



- All 12 recordings use a single channel sampled at 256 Hz, so statistics are directly comparable across sessions.
- Recording duration varies considerably, from about 1h 11m (2026-04-11) to about 22h 27m (2026-03-25).
- The 2026-06-02 and 2026-06-29 recordings have matching percentile values and near-identical histogram shapes, suggesting very similar signal characteristics or a data duplication between the two sessions.
- Standard deviation of amplitude ranges from about 82.6 uV (2026-01-03) to about 303.1 uV (2026-05-05), indicating substantially different signal variability/noise levels between sessions.
- Peak-to-peak amplitude (max minus min) ranges from roughly 2,483 uV (2026-05-25) to about 3,577 uV (2026-03-25).

This report is derived strictly from the statistical fields present in the source dataset. It does not represent a clinical or diagnostic ECG interpretation. Any assessment of cardiac rhythm, heart rate, or abnormality requires review of the underlying waveform by a qualified clinician or an appropriately validated diagnostic algorithm, neither of which is reflected in this report.

11 CBC / Labs

12 reports, 01 Jan – 30 Jun 2026

Overall hematological status: Stable within reference ranges. No CBC parameter abnormality was identified against the report-provided reference ranges across the observation period.

PARAMETER	FIRST	LATEST	RANGE	TREND
Hemoglobin (Hb)	14.4	15.1	13.5 – 15.9	increasing
Total RBC count	5.0	5.1	4.8 – 5.3	increasing
Packed Cell Volume (PCV)	44.7	43.5	42.2 – 48.4	decreasing
Mean Corpuscular Volume (MCV)	90.0	97.8	88.4 – 98.8	increasing
MCH	27.6	29.9	27.6 – 31.1	increasing
MCHC	33.1	33.0	32.9 – 34.3	decreasing
RDW	12.1	13.5	12.1 – 13.5	increasing
Total WBC count	5761.0	9630.0	5476.0 – 9973.0	increasing
Neutrophils	58.0	54.0	52.0 – 61.0	decreasing
Lymphocytes	22.0	29.0	22.0 – 34.0	increasing
Eosinophils	4.0	4.0	1.0 – 5.0	stable
Monocytes	3.0	3.0	1.0 – 8.0	stable
Basophils	1.0	1.0	0.0 – 2.0	stable
Platelet Count	251725.0	209427.0	189551.0 – 373102.0	decreasing

Patient-reported vs. measured discrepancy: the 30 Jun 2026 questionnaire records patient recollections of White Blood Cell (WBC) count baseline trend from your latest Complete Blood Count (CBC): Mild Leukopenia (Borderline low); Platelet Count profile stability from recent blood evaluations: Borderline Thrombocytopenia (Low); Liver Function Test (LFT) - SGOT/AST status: Mild Elevation (40-100 U/L); measured CBC values across all 12 reports in this period remained within their reference ranges throughout (see table below). The discrepancy likely reflects patient recall/perception rather than a measured lab abnormality.

12 Sleep, Cognition & Quality of Life

Sleep

TREND	Sleep remained variable across visits
AVERAGE	6.2 hours per night across printed assessments

Daytime naps needed
Difficulty falling back asleep
Fatigue level through the day
Waking up unrefreshed

Cognition

MEMORY	Stable on clinician review, with low to intermittent patient-reported memory difficulty
ATTENTION	Variable patient-reported concentration and attention symptoms
LANGUAGE	Mild word-finding difficulty was noted but not described as worsening
MENTAL CLARITY	Improved mental clarity was reported at the latest visit

Quality of Life

DAILY FUNCTION	Independence in daily activities was repeatedly normal, intact, or stable
SOCIAL FUNCTION	Intermittent social withdrawal or isolation symptoms were reported
CLINICAL INTERPRETATION	Quality of life remains affected by seizures, sleep, triggers, and medication tolerability, despite generally stable outpatient status.

Latest Questionnaire Domain Scores (0-100)



13 Digital Twin Risk Dashboard

Overall clinical risk: Moderate

Future Seizure Risk: Moderate	Status Epilepticus Risk: Low To Moderate	Hospitalization Risk: Low To Moderate	Cognitive Decline Risk: Low
Medication Toxicity Risk: Low To Moderate	Functional Decline Risk: Low	EEG-Derived Seizure Risk: Elevated	Genetic Safety Flags: 10 Flags, 7 High

State Vector (Health Metric Scores, 0-100)



Each domain score is the mean of clinician-documented per-visit ratings across the clinical-note window; overall health score is the mean of the other domains. Derived digital-twin summary metrics, not a direct lab/imaging measurement.

14 Clinical Action Plan

Priority Actions (Pharmacogenomics)

HIGH **Avoid or use with heightened monitoring**

See priority_safety_flags for the specific gene-drug pairs (ABCB1 (CC), BDNF (TC), CREB1 (AA), FKBP5 (TC), GRIA3 (A), HTR1B (TA), METTL21A (AA)).

MODERATE **Dose with caution / consider therapeutic drug monitoring**

See metabolizer_profile for reduced-function genes (CYP2C9, CYP2B6, CYP3A5, UGT1A4, POR).

INFORMATIONAL **Contextualize**

Interpret alongside actual clinical history, current medication list, and labs -- this panel alone does not establish a diagnosis.

Recommended Actions

- continue the documented antiseizure medication regimen unless changed by the treating clinician
- maintain the documented plan for consistent sleep schedule and trigger avoidance
- continue seizure and aura tracking across follow-up windows
- continue periodic CBC, liver function, and antiseizure drug-level safety monitoring when scheduled
- reassess at the planned follow-up interval

Monitoring Priorities

- seizure frequency by type
- seizure clustering
- average and longest seizure duration
- post-ictal recovery time
- aura pattern and severity
- sleep duration and sleep fragmentation
- missed medication doses and trigger exposure
- dizziness and other medication side effects
- CBC, liver function, and antiseizure drug-level tracking
- word-finding, memory, and attention symptoms
- Signal Variance
- Amplitude Dynamics
- Electrical Stability
- Preictal Detection
- Seizure Frequency
- Temporal Disease Progression

MRI Follow-Up

- Perform formal radiology review of the MRI image volumes.
- Run automated brain segmentation and longitudinal registration across the three study identifiers.
- Perform hippocampal volumetry because epilepsy-related mesial temporal abnormalities cannot be excluded from metadata alone.
- Run FLAIR white matter lesion analysis if clinically relevant.
- Compare future MRI studies using the same T1w and FLAIR acquisition strategy when possible.
- Correlate MRI findings with seizure semiology and neurological examination.
- Review MRI together with EEG findings for seizure-focus localization.
- Repeat or escalate imaging review if new neurological deficits, worsening seizures, or new focal symptoms occur.

Medication Reconciliation

- Current regimen reconciled and checked against curated reference pharmacology and this patient's own pharmacogenomic findings (Section 07). Predicted-model (SuperCYPsPred) coverage is still unresolved, so a live, comprehensive DDI screen is recommended before adding or changing any interacting medication.